

DECLARATION

We, Benjamin Fredy Nkane, Said Haruna Shabani, and Fadhili Michael Clever, students in the Department of Computer Science at Ruaha Catholic University (RUCU), Iringa, hereby declare that this project report titled *Track My Device with AI Assistant* is our original work. This project focuses on the design and development of a mobile-based device tracking system that integrates GPS, AI-based movement prediction, geofencing, and anti-theft alerts to improve the security and recovery of mobile devices. No part of this work has been submitted for any academic degree or award at this university or elsewhere. All external sources used in this report have been properly cited and referenced.

BENJAMIN FREDY NKANE

RU/BCS/2023/198

.....

Signature

.....

Date

SAID HARUNA SHABANI

RU/BCS/2023/197

.....

Signature

.....

Date

FADHILI MICHAEL CLEVER

RU/BCS/2023/114

.....

Signature

.....

Date

CERTIFICATE

This is to certify that the project report titled *Track My Device with AI Assistant* submitted by Benjamin Fredy Nkane, Said Haruna Shabani, and Fadhili Michael Clever to the Department of Computer Science, Ruaha Catholic University (RUCU), Iringa, is a record of their original research carried out under my supervision during the academic year 2025/2026. It is affirmed that this work has not been previously submitted for any other academic qualification.

.....

Mr. BENEDICT LOGATHO.

.....

Date.

ACKNOWLEDGEMENT

First and foremost, we would like to express our sincere gratitude to Almighty God for His guidance, strength, and blessings throughout the process of completing this project. His grace has enabled us to overcome challenges and successfully accomplish this academic work.

We would like to extend our deepest appreciation to our supervisor, Mr. Benedict Logatho, for accepting the responsibility of supervising this project. We are highly grateful for his valuable guidance, professional advice, constructive feedback, and continuous support throughout the preparation and completion of this project. His commitment and dedication have significantly contributed to the successful achievement of this work.

Our sincere appreciation goes to the Department of Computer Science at Ruaha Catholic University (RUCU), Iringa, for providing the necessary facilities, academic resources, and technical environment that enabled us to successfully carry out this project.

We are deeply thankful to our families for their love, prayers, and understanding during this demanding academic period. We also acknowledge our friends and colleagues for their encouragement and moral support throughout the journey.

We further appreciate the participants who volunteered to test the system and provided constructive feedback, which greatly contributed to the improvement and usability of the application.

We also extend our thanks to all individuals who offered assistance, guidance, or encouragement in any form. Your support is sincerely acknowledged and appreciated.

ABSTRACT

The increasing reliance on mobile devices such as smartphones, tablets, and laptops has intensified concerns regarding device security, tracking, and recovery, as these gadgets store vast amounts of personal, financial, and professional information that can be exploited if lost or stolen. Existing tracking solutions often provide only basic location services with limited predictive capabilities, lack real-time intelligence, and operate as fragmented standalone applications that fail to offer comprehensive protection against threats such as SIM card replacement, device shutdown, unauthorized movement, and low battery conditions.

This project presents the design and development of a *Track My Device with AI Assistant system*, a unified mobile-based and web based platform that integrates real-time GPS tracking, machine learning-based movement prediction, geofencing alerts, SIM change notifications, low battery warnings, remote lock, alarm activation, and an AI-powered chatbot into a single cohesive solution.

The system was developed using a Design Science Research (DSR) methodology to ensure a structured approach to problem identification, solution design, and evaluation. The backend was developed using the Laravel Framework with PHP 8.4 and hosted on a LiteSpeed web server for optimized performance. The user interface was designed using Bootstrap 5 and Blade Templates, with secure authentication implemented using Cookie Sessions and CSRF tokens. A RESTful JSON API was developed to enable seamless communication between the Android application and the server.

The system was tested through simulations and user acceptance testing, demonstrating reliable tracking accuracy, timely alert delivery, and high user satisfaction. The multi-device management dashboard allows users to monitor all registered devices from a single interface, displaying real-time location, battery status, and security alerts.

Overall, the system provides an intelligent, integrated, and user-friendly solution for mobile device protection, particularly in regions where access to advanced security technologies is limited.

TABLE OF CONTENTS

DECLARATION.....	i
CERTIFICATE.....	ii
ACKNOWLEDGEMENT.....	iii
ABSTRACT.....	iv
TABLE OF CONTENTS	v
LIST OF FIGURES	vii
LIST OF TABLES	ix
LIST OF ABBREVIATIONS	x
CHAPTER ONE	1
INTRODUCTION.....	1
1.1 Project Overview.....	1
1.2 Problem Statement.....	2
1.3 Objectives.....	3
1.4 Scope of the Project.....	3
1.5 Significance of the Study	4
1.6 Project Organization.....	4
CHAPTER TWO	6
LITERATURE REVIEW	6
2.1 Existing Systems.....	6
2.3 Proposed System	8
2.4 Technology Review	8
CHAPTER THREE.....	10
REQUIREMENT ANALYSIS.....	10
3.1 Requirement Elicitation Methods.....	10
3.2 Functional Requirements	10
3.3 Non-Functional Requirements.....	11
3.4 System Requirements	12
3.5 Use Case Modeling.....	12
CHAPTER FOUR.....	17
SYSTEM DESIGN.....	17
4.1 System Architecture.....	17

4.2 UML Diagrams.....	18
4.3 Database Design	23
4.4 UI/UX Design / Wireframes / Mockups	26
4.5 API Design	27
CHAPTER FIVE	28
IMPLEMENTATION	28
5.1 Development Methodology.....	28
5.2 Technology Stack	28
5.3 Key Modules and Features Implementation.....	29
5.4 Challenges.....	35
CHAPTER SIX	36
TESTING AND QUALITY ASSUARANCE	36
6.1 Testing Strategy	36
6.2 Types of Testing Done.....	36
6.3 Test Cases Table.....	37
6.4 Bug Tracking.....	38
CHAPTER SEVEN.....	39
RESULTS AND DISCUSSION	39
7.1 System Output.....	39
7.2 Performance Evaluation.....	44
7.3 Limitations.....	44
CHAPTER EIGHT.....	45
CONLUSION & FUTURE WORK.....	45
8.1 Summary of Work	45
8.2 Contribution	46
8.3 Future Enhancements.....	47
REFERENCES.....	48
APPENDECIES	49
Appendix 1: Source Code.....	49
Appendix 2: Project timeline and expenses	50
Project Timeline	50
Project Expenses	50

LIST OF FIGURES

Figure 1: Use case diagram for the Track My Device with AI Assistant system showing the interactions between the User, AI Chatbot, System, and the nine core use cases including User Registration, Device Registration, AI Chatbot Assistance, AI Prediction, Intelligent Alerts, Route Option, Send Command And Geofencing.	16
Figure 2: System architecture of the Track My Device with AI Assistant.	17
Figure 3: Class diagram for the Smart Agro Intelligent System showing the main classes, their attributes, methods, and relationships.	18
Figure 4: Sequence diagram for the device registration feature in the Track My Device with AI Assistant system.	19
Figure 5: Sequence diagram for the AI chatbot assistance feature in the Track My Device with AI Assistant system.	19
Figure 6: Sequence diagram for the send command feature in the Track My Device with AI Assistant system.	20
Figure 7: Activity diagram for the device registration process in the Track My Device with AI Assistant system.	20
Figure 8: Activity diagram for the AI chatbot assistance process in the Track My Device with AI Assistant system.	21
Figure 9: Activity diagram for the send command process in the Track My Device with AI Assistant system.	22
Figure 10: Activity diagram for the geofencing alert generation process in the Track My Device with AI Assistant system.	22
Figure 11: Entity-Relationship (ER) diagram for the Track My Device with AI Assistant system database.	23
Figure 12: Relational Database Schema for the Track My Device with AI Assistant System	26
Figure 13: UI/UX Design of Track My Device with AI Assistant system	26
Figure 14: User registration page showing input fields for name, email, phone number, and password. ..	30
Figure 15: Device registration interface showing the unique code entry field for linking a new device....	30
Figure 16: Multi-device dashboard displaying registered devices on an interactive map with location pins and status indicators.	31
Figure 17: AI chatbot assistant interface showing conversation starter prompts and user interaction window.....	31
Figure 18: AI analysis interface displaying device status assessment and battery health prediction.	32
Figure 19: Alerts interface showing a list of low battery notifications for a registered device.	32
Figure 20: Geofencing interface showing geofence creation, editing, and status management options.	33
Figure 21: Send command interface showing available remote commands including lock, unlock, alarm, and GPS actions.	33
Figure 22: All devices list interface displaying registered devices with their status, battery level, and last seen timestamp.	34
Figure 23: Route options interface showing date and time selection for viewing device movement history.	34
Figure 24: Live Map Dashboard Provides real-time geospatial tracking of all registered devices, displaying exact coordinates, battery status, and network connectivity on a high-resolution interactive map.....	39

Figure 25: AI Assistant (ChatBot) Enables conversational device management and instant status queries through natural language commands.	40
Figure 26: AI Predictive Analysis Delivers deep insights into battery health, movement patterns, and system anomalies.	40
Figure 27: Alert Management Log Displays a structured timeline of all critical notifications and battery events in one place.	41
Figure 28: Route Navigation calculates the optimal turn-by-turn path to guide you directly to your lost device's location.	41
Figure 29: Remote Device Lock Instantly locks the screen and blocks all unauthorized access until an admin PIN is entered.	42
Figure 30: Remote Siren Alarm Triggers a full-screen red alert and loud siren for immediate auditory and visual recovery.	43

LIST OF TABLES

Table 1: Test Case Table.....	37
Table 2: Bugs Tracking Table.....	38

LIST OF ABBREVIATIONS

AI – Artificial Intelligence
GPS – Global Positioning System
IoT – Internet of Things
ICT – Information and Communication Technology
ML – Machine Learning
API – Application Programming Interface
DBMS – Database Management System
UI – User Interface
UX – User Experience
SDLC – Software Development Life Cycle
UML – Unified Modeling Language
SQL – Structured Query Language
LAN – Local Area Network
WAN – Wide Area Network
GSM – Global System for Mobile Communications
SMS – Short Message Service
RFID – Radio Frequency Identification
OS – Operating System
RAM – Random Access Memory
CPU – Central Processing Unit

CHAPTER ONE

INTRODUCTION

1.1 Project Overview

The increasing reliance on mobile devices such as smartphones, tablets, and laptops has transformed the way individuals communicate, work, store information, and conduct daily activities. These devices have become essential tools for personal, academic, and professional use, storing vast amounts of sensitive data including financial records, personal communications, academic documents, and professional correspondence [1]. However, the growing dependence on these devices has also intensified concerns regarding device security, tracking, and recovery, as their loss or theft can lead to serious consequences, including data exposure, financial loss, privacy violations, and disruption of communication [2]. In Tanzania, the rapid adoption of mobile technology has been accompanied by an increase in device theft and loss, particularly in urban areas and public transportation hubs, where incidents of smartphone theft are common [3].

Despite the availability of various tracking and security applications, most existing solutions provide only basic location services with limited predictive capabilities and fragmented functionality [4]. Many tracking systems rely solely on GPS technology to provide real-time location updates, but they often fail to address critical security events such as SIM card replacement, device shutdown, unauthorized movement, and low battery conditions [5]. Additionally, most of these solutions operate as standalone applications, requiring users to install multiple apps for different security functions, such as location tracking, remote locking, alarm activation, and anti-theft protection [6]. This fragmentation makes it difficult for users to effectively manage and secure their devices, particularly in emergency situations where quick response is essential [4].

The emergence of artificial intelligence (AI) and machine learning technologies has opened new possibilities for enhancing device tracking and security systems [7]. AI-powered systems can analyze movement patterns, predict future device locations, and generate intelligent alerts when suspicious behavior is detected [8]. For example, machine learning algorithms can use historical GPS data to predict the likely trajectory of a lost or stolen device, improving the chances of recovery [9]. Additionally, conversational AI and AI-powered chatbots have been shown to make digital systems more intuitive and accessible, enabling users to interact with tracking systems

through natural language rather than navigating complex menus [10]. These advancements offer the potential to address the limitations of existing tracking solutions by providing intelligent, integrated, and user-friendly platforms for device security and recovery.

In Tanzania, the growing smartphone penetration and increasing reliance on mobile devices for financial transactions, communication, and access to government services have heightened the need for effective device tracking and security solutions [3]. However, there is currently no comprehensive, locally adapted system that integrates real-time GPS tracking, AI-based movement prediction, intelligent alert mechanisms, and conversational AI into a single platform tailored to the needs of Tanzanian users [6]. Most existing solutions are designed for international markets and do not consider local challenges such as unreliable internet connectivity, limited access to customer support, and language barriers [5]. This gap underscores the need for a Track My Device with AI Assistant system that combines multiple security features into a unified, accessible, and intelligent platform, offering Tanzanian users a practical and reliable solution for protecting their mobile devices.

1.2 Problem Statement

Despite the growing dependence on mobile devices for communication, work, and data storage, users continue to face significant challenges in locating and securing lost or stolen devices. Existing tracking solutions provide limited functionality, often lacking real-time monitoring, predictive analysis, and adequate response to critical events such as SIM card changes, device shutdown, or unauthorized movement. Most systems operate as fragmented standalone tools, requiring users to switch between multiple applications to access location updates, alerts, and security controls, which reduces efficiency and delays emergency responses. Additionally, many solutions are not designed for weak network environments, making them unreliable in rural or remote areas, and they lack conversational interfaces that could simplify interaction for non-technical users. These shortcomings reduce recovery success rates and expose users to privacy and financial risks. To address this, the project proposes a Track My Device with AI Assistant system that integrates GPS tracking, machine learning-based movement prediction, geofencing, SIM change alerts, remote lock, alarm activation, and an AI chatbot into a single, intelligent, and user-friendly platform.

1.3 Objectives

1.3.1 Main objective

Main Objective To design and develop an AI-powered smart device tracking system that enables real-time location monitoring, intelligent prediction of device movement, secure alert mechanisms, and interactive user engagement through a Chatbot interface.

1.3.2 Specific objectives

- i. To develop a real-time device tracking system using GPS technology that ensures accurate location monitoring even in areas with weak network connectivity.
- ii. To design an AI-powered monitoring system that analyzes device history and uses techniques to predict the possible future status of device.
- iii. To implement an intelligent alert and security system that notifies users when the device leaves a defined geographical area, when the SIM card is changed, when the device is turned off, or when the battery level is low, while also providing anti-theft features such as remote lock and alarm, and integrating an AI Chatbot for conversational interaction.
- iv. To create a user-friendly multi-device management system that allows users to monitor multiple devices through an interactive dashboard displaying location, status, and battery information.

1.4 Scope of the Project

This project focuses on the design and development of a Track My Device with AI Assistant system, a unified mobile-based platform that integrates real-time GPS tracking, machine learning-based movement prediction, geofencing alerts, SIM change notifications, low battery warnings, remote lock, alarm activation, and an AI-powered chatbot into a single cohesive solution.

The system is developed using the Laravel Framework with PHP 8.4 for backend development, LiteSpeed as the web server, Bootstrap 5 and Blade Templates for the user interface, and RESTful JSON APIs for communication between the Android mobile application and the server.

The system targets individual users, families, and small organizations who own multiple mobile devices and seek an intelligent, integrated, and user-friendly solution for device monitoring and protection. The scope includes device registration, real-time location tracking, geofencing, anti-theft alert generation, chatbot-based user interaction, and multi-device management through a centralized dashboard. The system is tested through simulations and user acceptance testing with

participants from urban and suburban areas in Tanzania to evaluate its performance, accuracy, and usability.

1.5 Significance of the Study

This project is motivated by the growing need for effective and intelligent device tracking and security solutions in an era where mobile devices have become indispensable tools for communication, work, education, and personal information storage. The loss or theft of a mobile device can lead to serious consequences, including data exposure, financial loss, privacy violations, and disruption of daily activities. Existing tracking solutions are often fragmented, offering only basic location services without addressing critical events such as SIM card replacement, device shutdown, unauthorized movement, or low battery conditions.

Furthermore, many systems are not designed for weak network environments and lack conversational interfaces that could simplify user interaction. The Track My Device with AI Assistant system aims to address these gaps by providing an integrated, intelligent, and user-friendly platform that combines real-time GPS tracking, AI-based movement prediction, geofencing alerts, anti-theft mechanisms, and an AI chatbot into a single unified solution.

The system is particularly significant for users in developing regions like Tanzania, where access to advanced security technologies is limited and reliable device recovery is critical. By offering a practical, accessible, and intelligent solution, this project contributes to enhanced device security, improved recovery rates, and greater user confidence in adopting technology-driven security measures for everyday use.

1.6 Project Organization

This report is organized into eight chapters. Chapter 1 provides the introduction, including the background, problem statement, objectives, scope, motivation, and organization of the project. Chapter 2 presents a review of related literature on device tracking technologies, AI-powered tracking systems, GPS-based monitoring, intelligent alert mechanisms, machine learning for movement prediction, AI chatbots, and multi-device management dashboards. Chapter 3 describes the requirement analysis, including elicitation methods, functional and non-functional requirements, system requirements, and use case modeling. Chapter 4 covers the system design, including system architecture, UML diagrams, database design, UI/UX wireframes, and API design. Chapter 5 discusses the implementation process, including the development methodology,

technology stack, key modules, and challenges encountered. Chapter 6 presents the testing and quality assurance strategies, including test cases, types of testing, and bug tracking results. Chapter 7 presents the results and discussion, including system screenshots, performance evaluation, and limitations. Finally, Chapter 8 concludes the report with a summary of the work, contributions, and recommendations for future enhancements.

CHAPTER TWO

LITERATURE REVIEW

2.1 Existing Systems

2.1.1 GPS and GSM-Based Real-Time Tracking Systems

Several existing systems use GPS and GSM technologies to provide real-time location tracking for mobile devices and vehicles. These systems typically rely on GPS satellites to determine the device's coordinates and GSM networks to transmit location data to a central server or the user's mobile phone. Sahoo and Rath (2012) developed a prototype model for location tracking and monitoring using GPS and GSM technologies, which allowed users to track the movement path of an object and receive location information via SMS on their mobile phones [1]. Similarly, Saghaei (2016) designed and implemented a fleet management system that used a GPS/GLONASS tracker and web-based software to provide real-time vehicle location, fuel consumption monitoring, and speed limit alerts, demonstrating the effectiveness of satellite-based tracking for fleet operations [2]. Jadhav et al. (2025) introduced an AI Guide Dog system that uses smartphone sensors and machine learning to predict egocentric paths, enabling real-time navigation assistance and movement tracking [3]. While these systems provide reliable location information, they often depend on continuous internet connectivity and may experience delays in areas with poor network coverage, which limits their effectiveness in rural or remote environments.

2.1.2 AI-Based Anti-Theft and Alert Systems

Intelligent anti-theft systems have been developed to detect and respond to unauthorized device movement, SIM card changes, and other suspicious activities. Wang et al. (2024) introduced ChatTracer, an LLM-powered real-time Bluetooth device tracking system that uses wireless sensor networks and large language models to provide intelligent tracking and user interaction [4]. The system demonstrated that LLMs can be effectively used for real-time device tracking and alert generation. Chang et al. (2021) developed CoSafe, a system that secures mobile devices through mutual mobility consistency verification, which detects theft by comparing the mobility patterns of multiple devices carried by the same user [5]. The system was able to detect stolen devices by identifying inconsistencies in movement patterns between paired devices. Wang et al. (2022) proposed GraphTrack, a graph-based cross-device tracking framework that models device relationships to detect suspicious activities and generate alerts [6]. While these systems offer enhanced security features, they often require multiple devices or additional hardware, and may not be suitable for individual users with a single device.

2.1.3 Multi-Device Management and Tracking Platforms

Multi-device management platforms provide centralized monitoring and control for multiple devices, enabling users to track location, enforce security policies, and remotely lock or wipe devices. Zhong et al. (2024) developed a trajectory smoothing system using GNSS/PDR integration via factor graph optimization, which improved the accuracy of device tracking in urban canyon environments where GPS signals are weak or obstructed [7]. The system demonstrated the importance of multi-sensor fusion for reliable device tracking in challenging environments. Jouaber et al. (2021) proposed a Neural Network Adapted Kalman Filter (NNAKF) for target tracking, which uses a recurrent neural network to dynamically adjust the process noise covariance matrix, improving tracking accuracy during sudden movement changes [8]. These multi-device management platforms are valuable for organizations and families managing multiple devices, but they often require significant infrastructure and may not be accessible to individual users seeking personal device protection.

2.2 Limitations of Existing Systems

Despite the advancements in GPS-based tracking, AI-based movement prediction, and anti-theft alert systems, several limitations continue to hinder their effectiveness and widespread adoption. GPS-based tracking systems often rely heavily on continuous internet connectivity and third-party infrastructure, making them unreliable in areas with poor network coverage, and their accuracy is highly dependent on satellite visibility and environmental factors such as urban canyons or indoor environments [1][2][3].

AI-based movement prediction systems, while offering enhanced predictive capabilities, require substantial training data and computational resources, making them difficult to deploy on low-power devices, and they often struggle with sudden, unexpected movements that deviate from historical patterns [8][9][11]. Anti-theft and alert systems frequently require multiple devices or additional hardware, making them inaccessible to individual users with a single device, and they may generate false alarms that reduce user trust and lead to alert fatigue [4][5][6]. Furthermore, many of these systems operate as standalone applications without integration with other security or management tools, forcing users to rely on multiple fragmented solutions to address their tracking and security needs. These limitations underscore the need for an integrated, intelligent, and accessible platform that combines real-time tracking, predictive analysis, anti-theft alerts, and conversational interaction within a single unified system.

2.3 Proposed System

The Track My Device with AI Assistant system is proposed as an integrated mobile-based platform designed to address the limitations of existing device tracking and security solutions by combining multiple functions into a single, intelligent, and user-friendly system. Unlike existing systems that operate as standalone applications offering only basic location services or isolated security features, the proposed system integrates real-time GPS tracking, AI-based movement prediction using a Neural Network Adapted Kalman Filter (NNAKF), geofencing alerts, SIM change notifications, low battery warnings, remote lock, alarm activation, and an AI-powered chatbot into one unified platform.

The system is developed using the Laravel Framework with PHP 8.4 for backend development, LiteSpeed as the web server, Bootstrap 5 and Blade Templates for the user interface, and RESTful JSON APIs for communication between the Android mobile application and the server. It targets individual users, families, and small organizations who own multiple mobile devices and seek a comprehensive solution for device monitoring and protection. By integrating AI-driven predictive capabilities, real-time alerts, and conversational interaction, the proposed system enables users to monitor their devices, receive timely notifications about security threats, and interact with the system through natural language queries, ultimately providing a practical, accessible, and intelligent solution for mobile device protection.

2.4 Technology Review

The development of the Track My Device with AI Assistant system is supported by a robust and secure technology stack designed to ensure reliability, performance, and user-friendliness. The backend is developed using the Laravel Framework with PHP 8.4, which provides a secure and scalable foundation for handling user authentication, device registration, location updates, alert generation, and API communication. The system is hosted on a LiteSpeed web server, which offers optimized performance and faster request handling, ensuring that the system can handle multiple concurrent users and real-time data exchanges efficiently.

The user interface is designed using Bootstrap 5 and Blade Templates, providing a responsive, modern, and user-friendly experience across different devices, including smartphones, tablets, and desktop computers. Secure authentication is implemented using Cookie Sessions and CSRF tokens, ensuring that user data and device information are protected from unauthorized access. Communication between the Android mobile application and the server is facilitated through

RESTful JSON APIs, enabling real-time data exchange for location updates, device status, and user commands. The system also utilizes a relational database model consisting of tables for Users, Devices, Location Updates, and Status Updates to manage device tracking information efficiently. This technology stack ensures that the system is secure, scalable, and capable of delivering real-time tracking and security features to users.

CHAPTER THREE

REQUIREMENT ANALYSIS

3.1 Requirement Elicitation Methods

Requirements for the Track My Device with AI Assistant system were gathered through a combination of literature review, analysis of existing device tracking solutions, and informal consultations with potential users. The literature review focused on identifying common challenges in mobile device security, tracking, and recovery, as well as the limitations of existing tracking systems. Analysis of existing solutions such as Apple's Find My, Google's Find My Device, and various third-party anti-theft applications helped to identify gaps in functionality, including weak predictive capabilities, fragmented features, and lack of conversational interfaces.

Informal consultations with mobile device users, including students, professionals, and small business owners, provided insights into user expectations, preferences, and pain points related to device tracking and security. The gathered requirements were then analyzed, categorized, and prioritized to ensure that the proposed system addresses the most critical needs of users, particularly in environments with limited internet connectivity and diverse device types.

3.2 Functional Requirements

The Track My Device with AI Assistant system fulfills the following functional requirements based on the needs of users seeking reliable and intelligent device tracking and security solutions:

1. **User Registration and Login:** The system allows users to register by providing their name, email address, and password. After registration, users can log in using their email and password to access all system features.
2. **Device Registration:** The system allows registered users to add devices by entering a unique code generated from the mobile application installed on the device. This code links the device to the user's account, enabling tracking and management.
3. **AI Chatbot Assistance:** The system provides an AI-powered chatbot that enables users to interact with the system through natural language queries. Users can ask questions such as "Where is my device?" or "What is my battery percentage?" and receive real-time responses about device status, location, and other relevant information.
4. **AI Prediction:** The system uses machine learning (Neural Network Adapted Kalman Filter – NNAKF) to analyze device movement history and predict the possible future

location of the device. This feature also provides predictive insights into device status, helping users anticipate potential security risks or movement patterns.

5. **Intelligent Alerts:** The system automatically generates alerts for critical events, including:
 - Low battery: Notifies the user when the device battery level falls below a configured threshold.
 - SIM card change: Alerts the user if the SIM card in the device is replaced.
 - Device offline: Notifies the user when the device is turned off or loses internet connectivity.
6. **Route Option (History Tracking):** The system allows users to view the historical movement path of a device by selecting a specific date and time. This feature enables users to see where the device has passed throughout the day, helping them track its movement patterns over time.
7. **Send Command Feature:** The system allows users to send remote commands to the device, including:
 - Send Alarm – Triggers an audible alarm on the device.
 - Turn Off Alarm – Stops the alarm on the device.
 - Lock Device – Remotely locks the device to prevent unauthorized access.
 - Unlock Device – Remotely unlocks the device.
 - Force GPS Location – Forces the device to send its current GPS location immediately.
8. **Geofencing:** The system allows users to define a virtual boundary (geofence) for each device. If the device crosses this boundary, the system automatically triggers an alert, notifying the user that the device has left the safe area.

3.3 Non-Functional Requirements

The Track My Device with AI Assistant system meets the following non-functional requirements to ensure usability, reliability, and performance.

The system is accessible via mobile devices, laptop, and tablets ensuring that users can track and manage their devices anytime, anywhere. The system provides a user-friendly interface with

intuitive navigation, clear labels, and easy-to-understand instructions, requiring minimal training for users to operate effectively.

The system ensures the security of user data, including device information, location histories, and personal details, through secure authentication, data encryption, and access control mechanisms. The system is available 24/7 with minimal downtime, ensuring that users can access tracking and security features whenever needed, particularly during emergencies.

The system is scalable to accommodate an increasing number of users and devices without significant performance degradation. The system responds to user requests within acceptable time limits, with location updates delivered within seconds and alerts triggered in real time.

The system is designed to be maintainable, with modular architecture, well-structured code, and clear documentation that allows for easy updates and future enhancements.

3.4 System Requirements

The Track My Device with AI Assistant system requires the following hardware and software components to operate effectively.

The backend is hosted on a LiteSpeed web server with the following minimum specifications: Intel Xeon or equivalent processor, 4GB RAM, 20GB storage, and a stable internet connection. The server supports PHP 8.4 and the Laravel Framework for backend development, and uses a relational database management system (MySQL) to store user data, device information, location updates, and alert logs.

The system is accessible via Android mobile devices running Android 8.0 or higher, with the mobile application developed to communicate with the server via RESTful JSON APIs. The system requires internet connectivity for real-time GPS tracking, alert generation, and AI chatbot functionality, though the Android application can cache some data for limited offline functionality. The system integrates with GPS services for location tracking, and uses third-party APIs for push notifications and SMS alerts.

3.5 Use Case Modeling

The Track My Device with AI Assistant system includes several use cases that represent the interactions between actors and the system. The primary actors include the User, the System Administrator, and the AI Chatbot. Key use cases include User Registration, Device Registration,

Real-Time GPS Tracking, Movement Prediction, Geofencing Alert Generation, SIM Change Notification, Remote Lock Activation, AI Chatbot Interaction, and Multi-Device Dashboard Access. The following are sample use case descriptions:

Use Case 1: User Registration

Actor: User

Precondition: The user has access to the mobile application and an internet connection.

Flow: The user opens the web based system, navigates to the registration page, and enters their name, email address, and password. The system validates the input, creates a user account, and grants the user access to the system.

Use Case 2: User Login

Actor: User

Precondition: The user is already registered.

Flow: The user opens the system, enters their email and password, and clicks the login button. The system validates the credentials, authenticates the user, and grants access to the dashboard.

Use Case 3: Device Registration

Actor: User

Precondition: The user is logged into the web based application.

Flow: The user navigates to the device registration page, enters the unique code generated from the mobile application installed on the device, and submits the form. The system validates the code, links the device to the user's account, and confirms successful registration.

Use Case 4: AI Chatbot Assistance

Actor: User, AI Chatbot

Precondition: The user is logged into the system and has at least one registered device.

Flow: The user opens the chatbot interface and types a question such as "Where is my device?" or "What is my battery percentage?" The system processes the query using natural language

understanding, retrieves the relevant device information from the database, and returns a conversational response with the requested details.

Use Case 5: AI Prediction

Actor: User, System

Precondition: The user is logged into web based system and has at least one registered device with historical movement data.

Flow: The system analyzes the device's movement history using the Neural Network Adapted Kalman Filter (NNAKF) and displays predictive insights about the device's future location and status. The user can view these predictions on the dashboard.

Use Case 6: Intelligent Alerts

Actor: System, User

Precondition: The user is logged into the system and has at least one registered device.

Flow: The system continuously monitors the device for critical events. When any of the following occurs, the system generates and sends an alert to the user:

- i. Low battery: The device battery level falls below a configured threshold.
- ii. SIM card change: The SIM card in the device is replaced.
- iii. Device offline: The device is turned off or loses internet connectivity.

Use Case 7: Route Option (History Tracking)

Actor: User

Precondition: The user is logged into the system and has at least one registered device with historical location data.

Flow: The user selects a specific date and time, and the system retrieves and displays the historical movement path of the device, showing where the device has passed throughout the selected day.

Use Case 8: Send Command

Actor: User

Precondition: The user is logged into the system and has at least one registered device.

Flow: The user selects the Send Command feature and chooses one of the following actions:

- i. Send Alarm: Triggers an audible alarm on the device.
- ii. Turn Off Alarm: Stops the alarm on the device.
- iii. Lock Device: Remotely locks the device.
- iv. Unlock Device: Remotely unlocks the device.
- v. Force GPS Location: Forces the device to send its current GPS location immediately.

Use Case 9: Geofencing

Actor: User, System

Precondition: The user is logged into the web based application and has at least one registered device.

Flow: The user defines a virtual boundary (geofence) for a specific device. The system continuously monitors the device's location. If the device crosses the defined boundary, the system automatically triggers an alert and notifies the user.

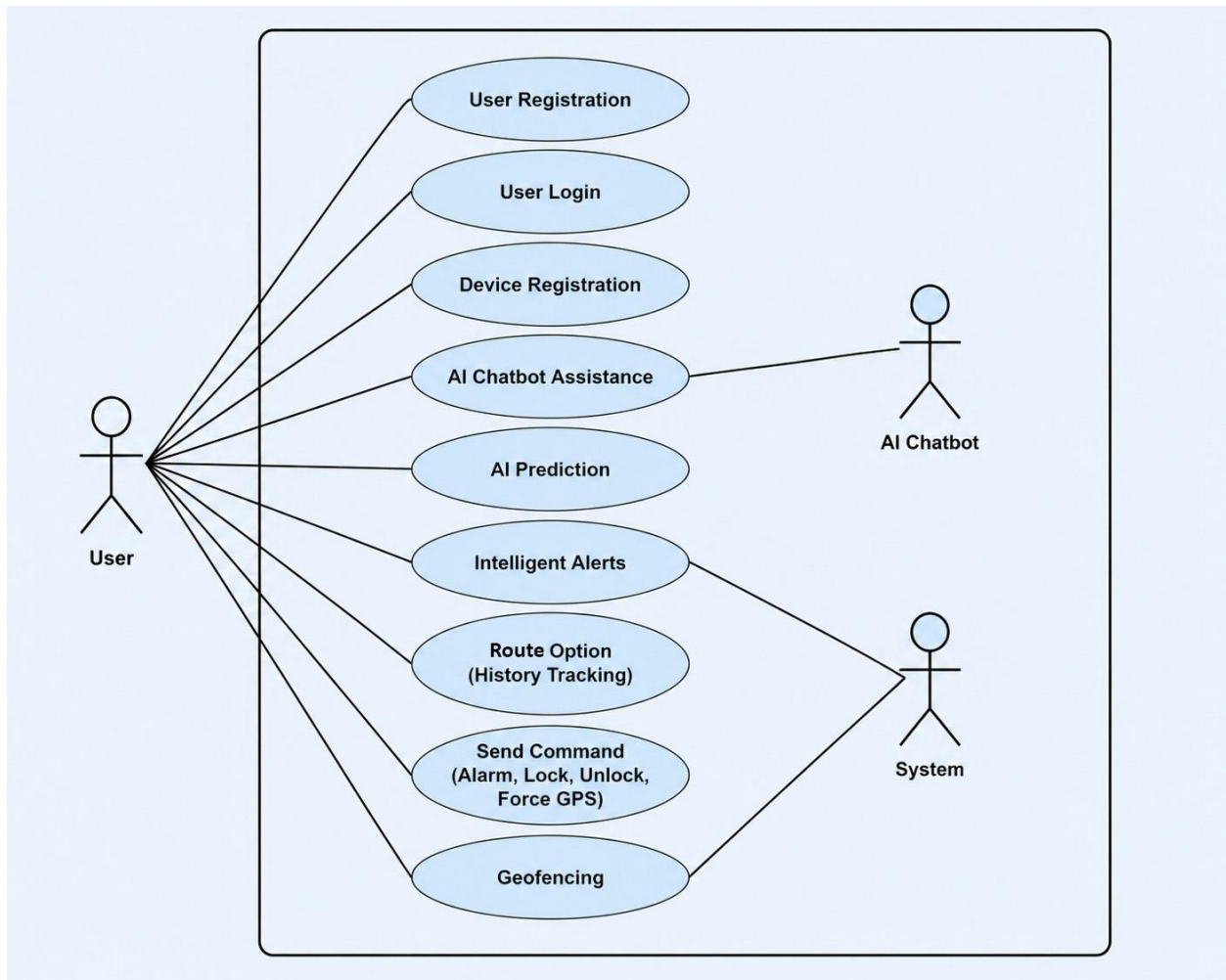


Figure 1. Use case diagram for the Track My Device with AI Assistant system showing the interactions between the User, AI Chatbot, System, and the nine core use cases including User Registration, Device Registration, AI Chatbot Assistance, AI Prediction, Intelligent Alerts, Route Option, Send Command And Geofencing.

CHAPTER FOUR

SYSTEM DESIGN

4.1 System Architecture

The Track My Device with AI Assistant system follows a client-server architecture with a three-tier design that separates the presentation layer, application logic, and data storage to ensure scalability, security, and efficient real-time data handling. The presentation layer is built using Bootstrap 5 and Blade Templates, providing a responsive and user-friendly interface that allows users to access the system through web browsers on both desktop and mobile devices.

The application layer is implemented using the Laravel Framework with PHP 8.4, which handles all server-side processing, including user authentication, device registration, location tracking, alert generation, and integration with external APIs for push notifications and SMS alerts.

The data layer consists of a MySQL relational database that stores and manages all application data, including user profiles, device information, location updates, and alert logs. The architecture also includes a RESTful JSON API that facilitates seamless communication between the Android mobile application and the server, enabling real-time data exchange for location updates, device status, and user commands.

The system is hosted on a LiteSpeed web server for optimized performance and faster request handling. The modular design ensures that the system is secure, scalable, and capable of delivering real-time tracking and security features to users, even in resource-constrained environments.

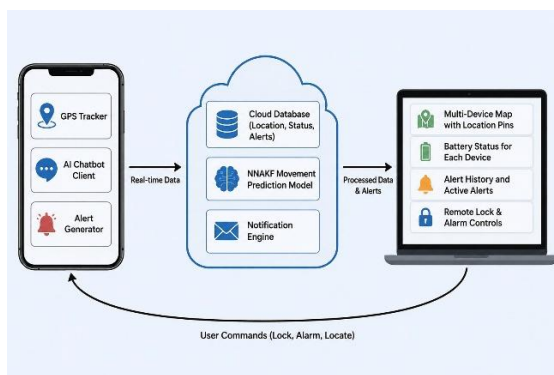


Figure 2: System architecture of the Track My Device with AI Assistant.

4.2 UML Diagrams

Unified Modeling Language (UML) diagrams were used to visually represent the structure and behavior of the Track My Device with AI Assistant system. The following diagrams were developed to provide a clear understanding of the system's design, interactions, and data flow.

4.2.1 Class Diagram

The class diagram represents the static structure of the system by showing the classes, their attributes, methods, and relationships. The main classes in the system include:

- **User:** Stores user details such as name, email, password, and registration date.
- **Device:** Contains information about registered devices, including device name, IMEI number, unique registration code, and registration date.
- **LocationUpdate:** Stores GPS location data for devices, including latitude, longitude, timestamp, and accuracy.
- **StatusUpdate:** Stores device status information, including battery level, connectivity status (online/offline), and last seen timestamp.
- **Alert:** Stores alert records, including alert type (low battery, SIM change, offline, geofence breach), timestamp, and status (read/unread).
- **Geofence:** Defines virtual boundaries for devices, including latitude, longitude, and radius.
- **Command:** Stores remote commands sent to devices, including command type (lock, unlock, alarm, force GPS), timestamp, and status (pending, sent, executed).
- **ChatLog:** Stores AI chatbot conversation history, including user queries, AI responses, and timestamps.

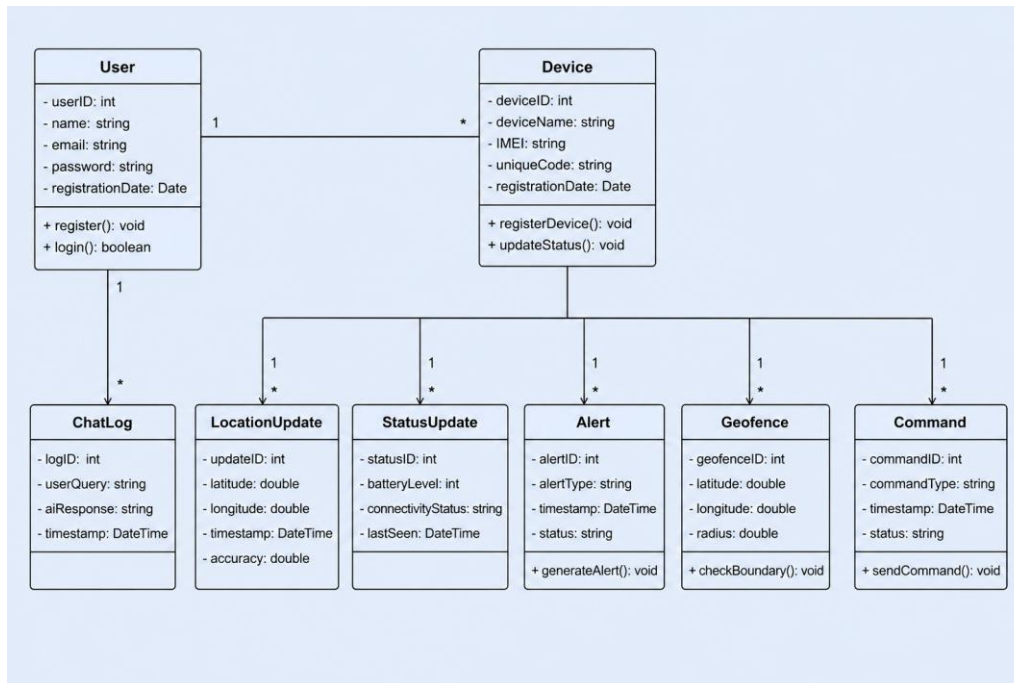


Figure 3: Class diagram for the Smart Agro Intelligent System showing the main classes, their attributes, methods, and relationships.

4.2.2 Sequence Diagrams

Sequence diagrams were used to model the interaction flow between actors and system components for key functionalities.

Sequence Diagram for Device Registration

The user logs into the mobile application, navigates to the device registration page, enters the unique code generated from the mobile application installed on the device, and submits the form. The system validates the code, links the device to the user's account, and confirms successful registration.

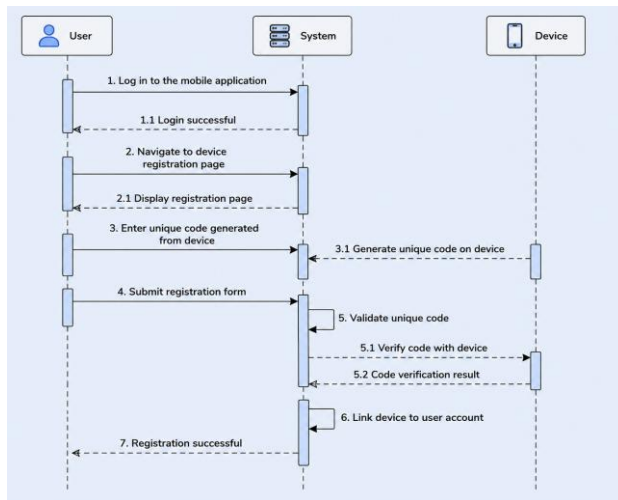


Figure 4: Sequence diagram for the device registration feature in the Track My Device with AI Assistant system.

Sequence Diagram for AI Chatbot Assistance

The user opens the chatbot interface and types a question (e.g., "Where is my device?"). The system processes the query using natural language understanding, retrieves the relevant device information from the database, and returns a conversational response with the requested details.

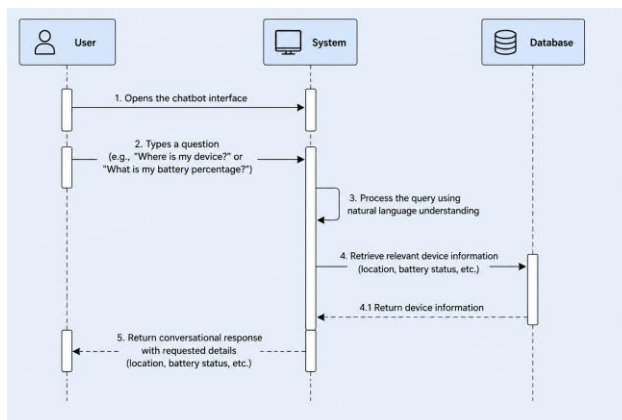


Figure 5: Sequence diagram for the AI chatbot assistance feature in the Track My Device with AI Assistant system.

Sequence Diagram for Send Command

The user selects the Send Command feature and chooses an action (e.g., Lock Device). The system sends the command to the target device via the RESTful API, receives confirmation, and updates the command status. A confirmation message is then returned to the user.

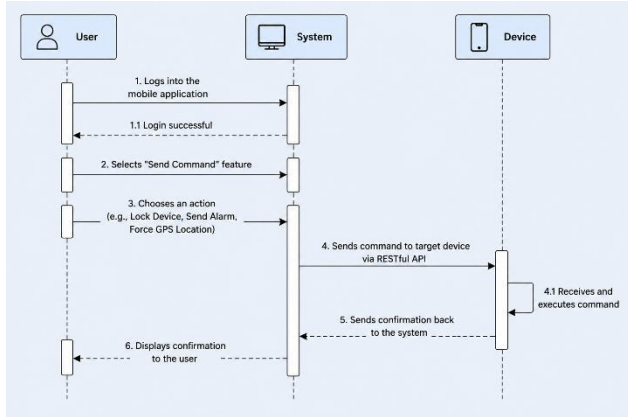


Figure 6: Sequence diagram for the send command feature in the Track My Device with AI Assistant system.

4.2.3 Activity Diagrams

Activity Diagram for Device Registration

The user logs into the mobile application and navigates to the device registration page. The user enters the unique code generated from the mobile application installed on the target device and submits it. The system validates the code against the database records. If valid, the device is linked to the user's account and a confirmation message is displayed. If invalid, an error message is shown and the user is prompted to re-enter the correct code.

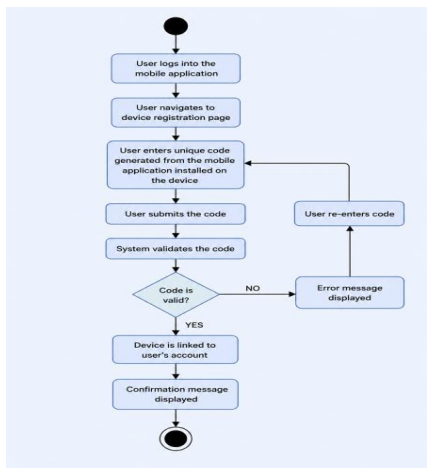


Figure 7: Activity diagram for the device registration process in the Track My Device with AI Assistant system.

Activity Diagram for AI Chatbot Assistance

The user opens the chatbot interface and types a natural language question about device status or location. The system processes the query using natural language understanding capabilities. The

system retrieves relevant device information from the database, such as location, battery level, or alert history. The system formulates a conversational response and returns it to the user through the chat interface. The process continues until the user closes the chatbot session.

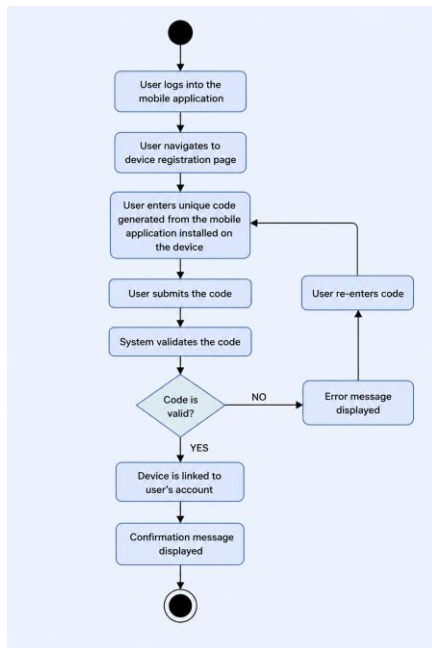


Figure 8: Activity diagram for the AI chatbot assistance process in the Track My Device with AI Assistant system.

Activity Diagram for Send Command

The user selects a registered device and chooses a command action from the available options: Lock, Unlock, Alarm, or Force GPS. The system processes the selected command and sends it to the target device through the RESTful API. The device receives and executes the command, then sends a confirmation response back to the system. The system updates the command status and displays a confirmation message to the user. If the command fails or the device is offline, an error message is displayed.

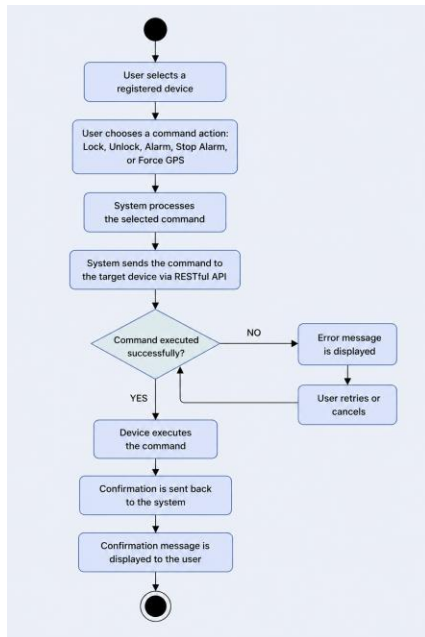


Figure 9: Activity diagram for the send command process in the Track My Device with AI Assistant system.

Activity Diagram for Geofencing Alert Generation

The system continuously monitors the real-time location of each registered device. For devices with active geofences, the system compares the device's current GPS coordinates with the geofence boundaries. If the device remains inside the geofence, monitoring continues without action. If the device crosses the boundary, an alert is triggered and recorded in the database. The user is immediately notified through the mobile application about the breach.

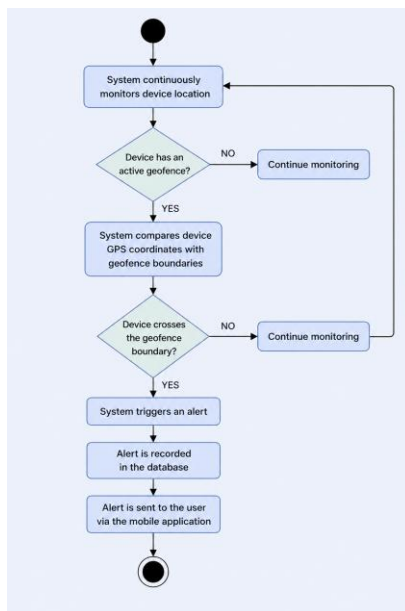


Figure 10: Activity diagram for the geofencing alert generation process in the Track My Device with AI Assistant system.

4.2.4 Entity-Relationship (ER) Diagram

The ER diagram represents the database structure and relationships between tables. The main entities include:

The relationships show that one user can have many devices, one device can have many location updates, status updates, alerts, commands, and geofences. One device can also have many chat logs associated with the user.

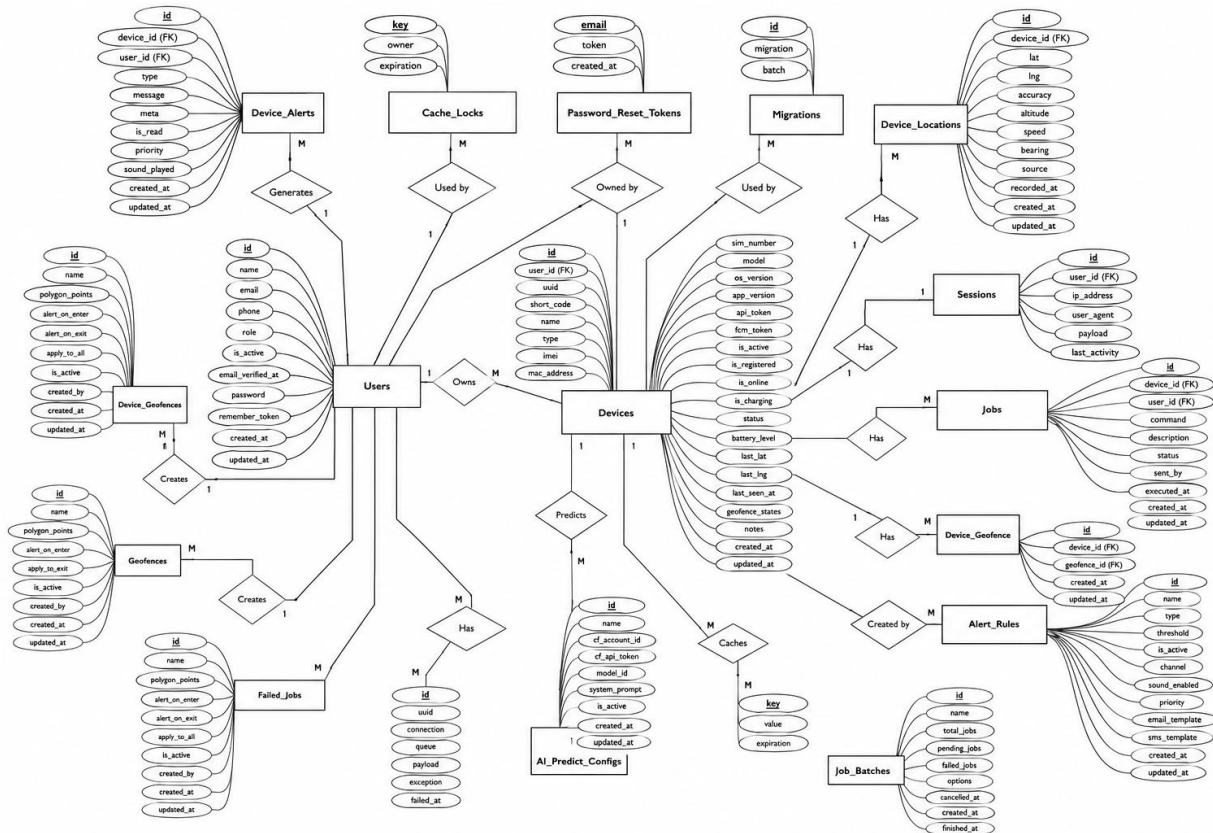


Figure 11: Entity-Relationship (ER) diagram for the Track My Device with AI Assistant system database.

4.3 Database Design

The system employs a comprehensive relational database architecture designed to manage the lifecycle of devices, streamline Artificial Intelligence (AI) prediction workflows, and handle asynchronous job processing. The schema is built on a normalized structure to ensure data consistency, facilitate complex queries, and maintain strict audit trails for device interactions. The database is divided into four distinct, interconnected operational modules:

User, Session, and Authentication Management

The core of the user-facing system resides in the `tmd_users` table. This table acts as the anchor for the database, storing essential user credentials including a unique email, a securely hashed password, and role-based access controls.

To support system scalability and security, the design includes a `tmd_sessions` table to actively track user login sessions, capturing IP addresses and user-agent data for security monitoring. Additionally, a `tmd_password_reset_tokens` table is provided to facilitate secure, time-limited password recovery operations.

Device Registry and Telemetry

Device management is centralized within the `tmd_devices` table, which stores comprehensive metadata for each registered unit, including hardware models, MAC addresses, IMEI numbers, and API tokens for authentication.

To monitor device health and mobility, the database employs a dedicated `tmd_device_locations` table. This table captures precise telemetry data (including latitude, longitude, altitude, speed, and bearing) in real-time, allowing the system to track device movement. Furthermore, a `tmd_device_geofence` table is utilized to link devices to virtual geographic boundaries, stored as polygon coordinates in the `tmd_geofences` table, enabling automated alerts when devices enter or exit specified zones.

Command Execution and Alerting

The system supports remote interaction with devices through a structured command and alerting framework. The `tmd_device_commands` table logs all directives sent from the backend to the devices (e.g., configuration updates or reboot instructions), tracking the command payload, its current status (pending, sent, completed), and the user who initiated it.

In parallel, the system proactively monitors device status and geofence breaches via the `tmd_device_alerts` and `tmd_alert_rules` tables. The `alert_rules` table allows administrators to define thresholds and conditions (such as timeouts or battery levels), while the `device_alerts` table logs the actual triggered incidents, storing the message, alert type, and priority level to ensure rapid response.

AI Prediction and Asynchronous Job Queues

A pivotal component of the architecture is the AI/ML integration framework, managed by the `tmd_ai_predict_configs` table. This table stores the configuration details for AI models, including API endpoints, secret keys, model paths, and system prompts, allowing the application to dynamically invoke prediction services.

To ensure the system remains responsive and scalable, all heavy computational tasks—such as data processing, AI predictions, or bulk operations—are handled asynchronously. This is achieved through the `tmd_jobs` and `tmd_job_batches` tables. These tables act as a distributed queue, storing pending jobs with their associated payloads, priority levels, and retry attempts. A `tmd_failed_jobs` table is included to log any exceptions or errors that occur during processing, ensuring system resilience and facilitating debugging.

System Optimization and Integrity

To maintain peak performance, the schema incorporates a `tmd_cache` and `tmd_cache_locks` table, which manage in-memory data caching and prevent race conditions during concurrent updates. Furthermore, a `tmd_migrations` table is included to version-control the database schema, allowing for seamless updates and rollbacks during the software development lifecycle. All major tables utilize standardized timestamp columns (`created_at`, `updated_at`) to maintain comprehensive historical records, and foreign key constraints are heavily utilized to preserve referential integrity across all device, user, and job-related tables.

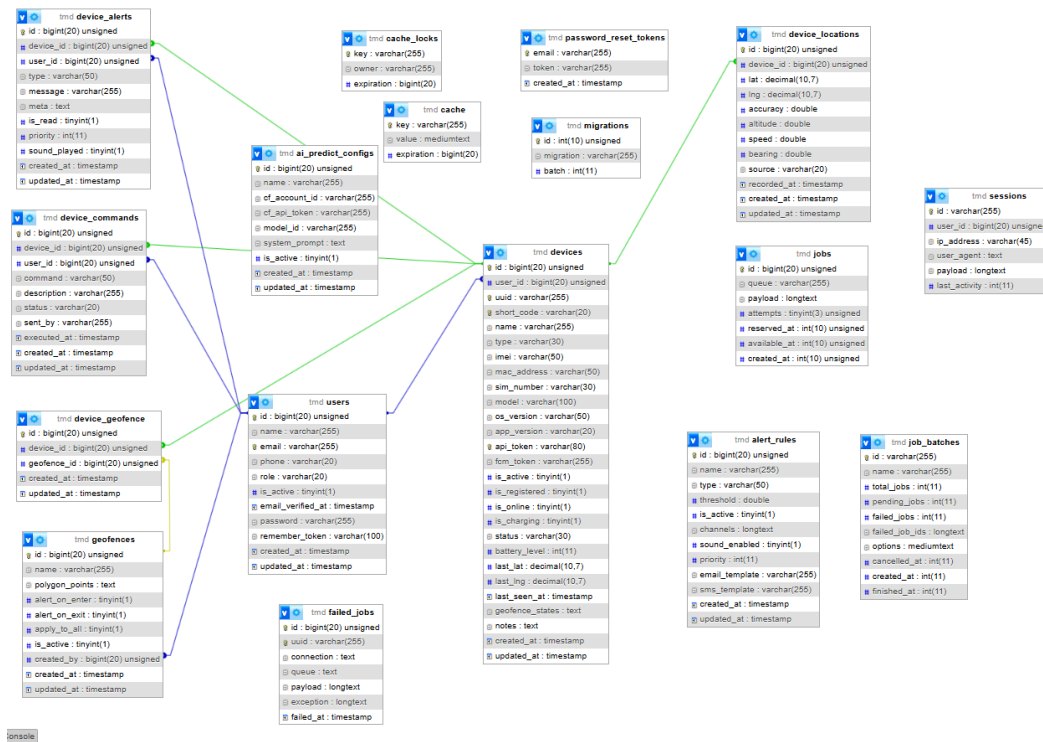


Figure 12: Relational Database Schema for the Track My Device with AI Assistant System

4.4 UI/UX Design / Wireframes / Mockups

The Track My Device with AI Assistant system features a modern, dark-themed dashboard designed for monitoring real-time data and remote device management. The interface is built with a strong emphasis on data visualization, allowing users to track device locations and performance metrics at a glance.

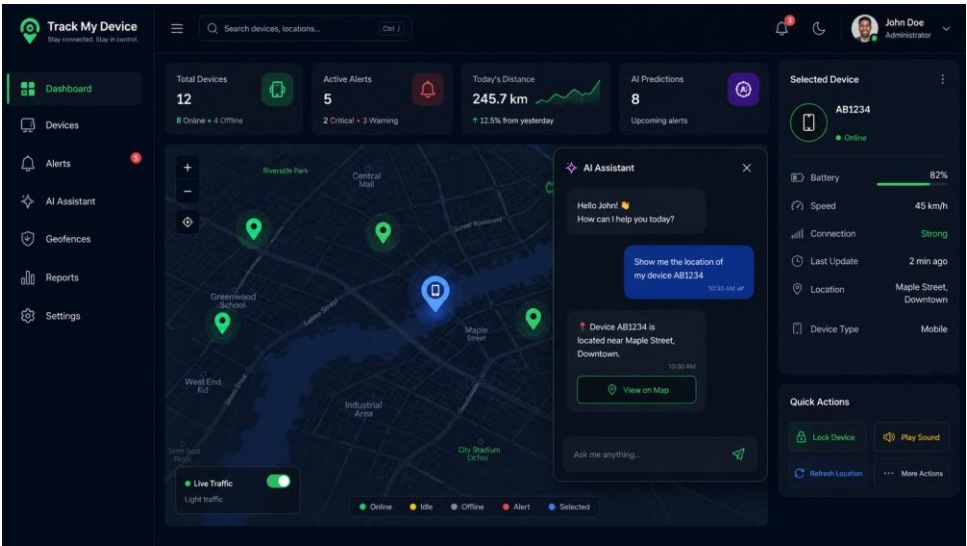


Figure 13: UI/UX Design of Track My Device with AI Assistant system

4.5 API Design

The TrackMyDevice API is a RESTful web service built on Laravel that enables real-time device tracking and monitoring through two distinct authentication mechanisms. Web users access the system through session-based authentication with CSRF protection, allowing them to view device dashboards and manage alerts via traditional web forms. Mobile applications, on the other hand, utilize device tokens for authentication, enabling secure data ingestion through specialized POST endpoints for location updates, status reports, and battery level monitoring. The API follows REST conventions with JSON data format over HTTPS, featuring a CORS policy that allows cross-origin requests from any domain, making it accessible across different platforms and applications. Key endpoints include `/api/devices` for retrieving device lists with real-time status, location coordinates, and battery levels, and `/api/alerts` for managing system notifications with priority levels, read status tracking, and multi-channel delivery support.

Core Functionality and Security Implementation

The system's dual-auth model effectively separates human user interactions from automated device communications, ensuring appropriate security for each access pattern. Mobile devices authenticate using bearer tokens to send GPS coordinates, connection status updates, and battery percentages through endpoints like `/api/device/location`, `/api/device/status`, and `/api/device/battery`, with failed authentication returning 401 errors. Web users authenticate through standard login forms, receiving session cookies that grant access to comprehensive dashboards and alert management interfaces. The alert system categorizes notifications by type (battery low, offline device, geofence breach) with priority levels from 1-5 and tracks unread counts, enabling efficient notification management. This architecture ensures secure, scalable device tracking with proper error handling, rate limiting, and data validation, making it suitable for applications requiring real-time monitoring of multiple devices with immediate alerting capabilities for critical events.

CHAPTER FIVE

IMPLEMENTATION

5.1 Development Methodology

The development of the TrackMyDevice system followed the Agile Software Development Methodology, which was selected for its ability to support iterative development, continuous testing, and regular improvement throughout the project lifecycle. Rather than building the entire system at once using a traditional waterfall approach, the project was divided into smaller, manageable modules that were designed, implemented, tested, and refined in successive iterations. This incremental approach allowed the development team to respond quickly to changing requirements, identify and fix issues early, and deliver working functionality at the end of each sprint. The Agile methodology proved particularly suitable for this project due to the complex integration requirements between the web interface, mobile device tracking, and real-time alert systems, where flexibility and rapid adaptation were essential.

Sprint Breakdown and Iterative Development

The development process was organized into two sprints, with each sprint focusing on specific functional areas of the system. The first sprint concentrated on setting up the Laravel backend infrastructure, database schema design, and implementing the dual authentication system for both web sessions and device tokens. Subsequent sprints addressed the development of core API endpoints for device and alert management, integration of the mobile device ingest endpoints for location and battery tracking, and finally the implementation of the frontend web interface. Each sprint began with planning sessions to define sprint goals, followed by daily stand-up meetings to track progress and address impediments, and concluded with review sessions to demonstrate working features and gather feedback for the next iteration. This structured yet flexible approach ensured continuous delivery of value, high software quality through frequent testing, and efficient feature integration that ultimately resulted in a robust, scalable device tracking system ready for deployment.

5.2 Technology Stack

The TrackMyDevice system was developed using a carefully selected technology stack that balances performance, security, and development efficiency. For the backend, Laravel Framework with PHP 8.4 was chosen as the primary development platform due to its robust ecosystem, built-in security features, and extensive support for RESTful API development. Laravel's Eloquent ORM provided seamless database interaction, while its authentication system enabled the implementation of dual authentication mechanisms for both web users and mobile devices. The application is served using LiteSpeed Web Server, which offers superior performance compared to traditional Apache servers, with features like built-in caching, HTTP/2 support, and efficient static file handling that significantly improve response times for API requests and web page delivery.

Frontend and Integration Technologies

For the user interface, Bootstrap 5 was integrated with Blade Templates to create responsive, server-side rendered views that provide a consistent and professional appearance across all

devices. This approach ensures that web users experience fast page loads and seamless navigation without the complexity of client-side JavaScript frameworks. Authentication is managed through Cookie Sessions combined with CSRF tokens, providing secure session management that protects against cross-site request forgery attacks while maintaining a smooth user experience.

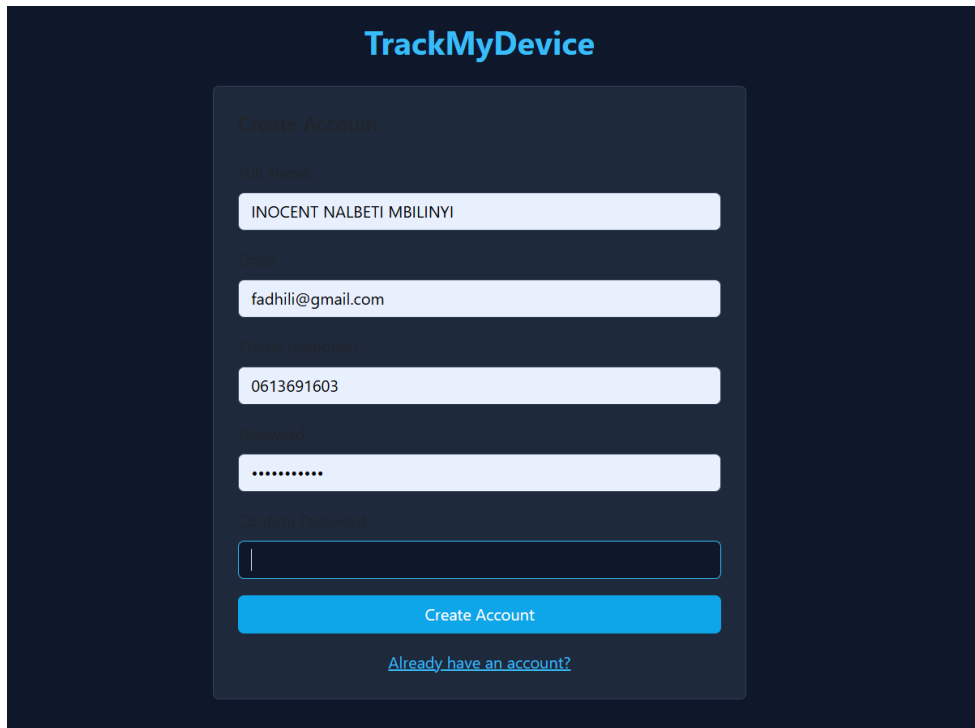
The system exposes RESTful JSON APIs that serve as the communication bridge between the Android tracking application and the server, enabling efficient data exchange for location updates, device status changes, and battery level monitoring. The underlying relational database model was designed with four core entities: Users managing device ownership, Devices storing tracking information, Location Updates recording GPS coordinates with timestamps, and Status Updates maintaining real-time device connectivity and battery status. This comprehensive technology stack ensures the system is secure, scalable, and capable of handling real-time device tracking requirements efficiently.

5.3 Key Modules and Features Implementation

The Track My Device with AI Assistant system consists of several key modules, each implemented to address specific user needs for device tracking and security.

5.3.1 User Registration and Authentication Module

The user registration module allows users to create accounts by providing their name, email address, and password. The registration form includes validation to ensure that all required fields are completed correctly. Upon successful registration, the system creates a new user record in the database and redirects the user to the login page. The authentication module handles user login by verifying the email and password against the database records. Passwords are stored securely using hashing algorithms. Once authenticated, the user is granted access to the system's features, including device registration, real-time tracking, chatbot assistance, and command execution. Session management ensures that users remain logged in until they explicitly log out.

The image shows a 'TrackMyDevice' user registration page. It features a dark blue background with a central white form titled 'Create Account'. The form contains five input fields: 'Full Name' (filled with 'INOCENT NALBETI MBILINYI'), 'Email' (filled with 'fadhili@gmail.com'), 'Phone (optional)' (filled with '0613691603'), 'Password' (filled with eight dots), and 'Confirm Password' (empty). Below the fields is a blue 'Create Account' button and a link that says 'Already have an account?'.

TrackMyDevice

Create Account

Full Name

INOCENT NALBETI MBILINYI

Email

fadhili@gmail.com

Phone (optional)

0613691603

Password

.....

Confirm Password

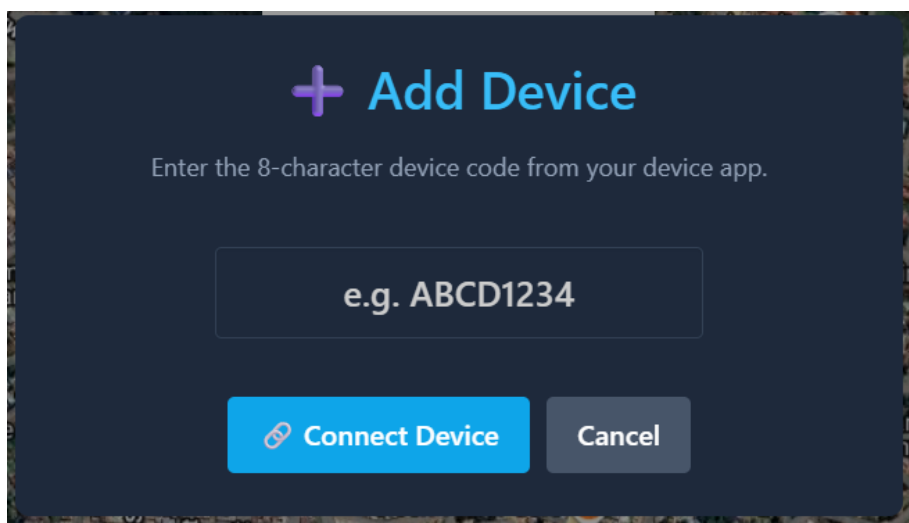
Create Account

[Already have an account?](#)

Figure 14: User registration page showing input fields for name, email, phone number, and password.

5.3.2 Device Registration Module

The device registration module allows users to add and manage their devices. Each device is registered using a unique code generated from the mobile application installed on the target device. The user enters this code into the web interface, and the system validates it against the database. Once validated, the device is linked to the user's account and becomes available for tracking and management. The module also allows users to update device information and view the status of registered devices.

The image shows the 'Add Device' interface. It has a dark blue background with a purple plus icon and the text 'Add Device'. Below this is a prompt: 'Enter the 8-character device code from your device app.' A large input field contains the example code 'e.g. ABCD1234'. At the bottom are two buttons: a blue 'Connect Device' button with a link icon and a grey 'Cancel' button.

+ Add Device

Enter the 8-character device code from your device app.

e.g. ABCD1234

Connect Device Cancel

Figure 15: Device registration interface showing the unique code entry field for linking a new device.

5.3.3 GPS Tracking and Location Monitoring Module

The GPS tracking module continuously captures the device's location using GPS technology and transmits the coordinates to the server at configurable intervals. The system stores the location data in the database and displays the device's current position on an interactive map within the user dashboard. The module also provides real-time location updates, allowing users to monitor their devices at any time. The GPS tracking feature is essential for locating lost or stolen devices.

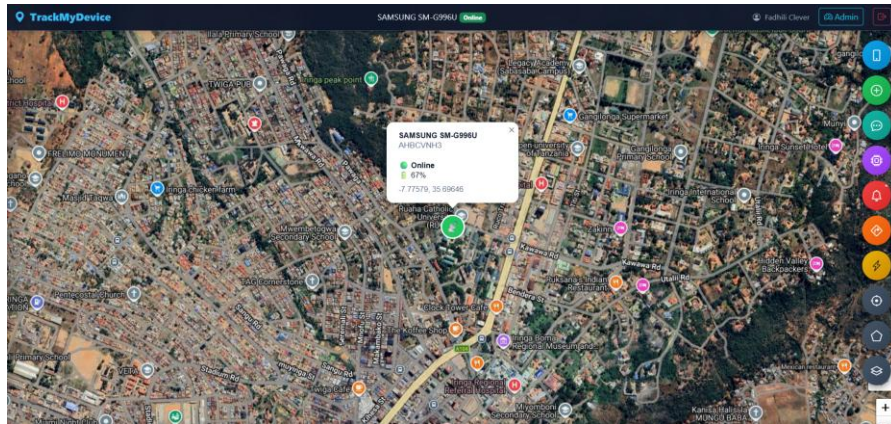


Figure 16: Multi-device dashboard displaying registered devices on an interactive map with location pins and status indicators.

5.3.4 AI Chatbot Assistant Module

The AI chatbot assistant module provides users with a conversational interface for interacting with the system. Users can ask questions about device status, location, battery level, and recent alerts. The module processes natural language queries and retrieves relevant device information from the database. The chatbot returns responses in a conversational format, making it easy for users to get quick answers without navigating complex menus.

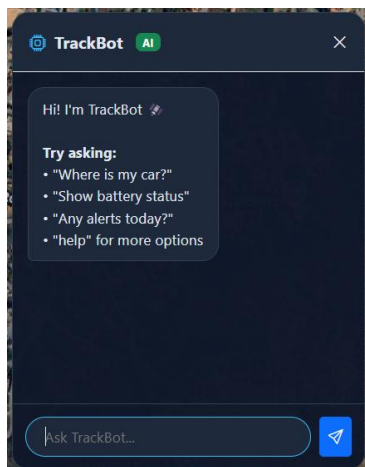


Figure 17: AI chatbot assistant interface showing conversation starter prompts and user interaction window.

5.3.5 Movement Prediction Module

The movement prediction module uses a Neural Network Adapted Kalman Filter (NNAKF) to analyze the device's historical movement data and predict its future location. The implementation involves collecting GPS trajectory data, processing it through the NNAKF algorithm, and displaying predictive insights to the user. This feature helps users anticipate where a lost device might be heading, improving the chances of recovery.

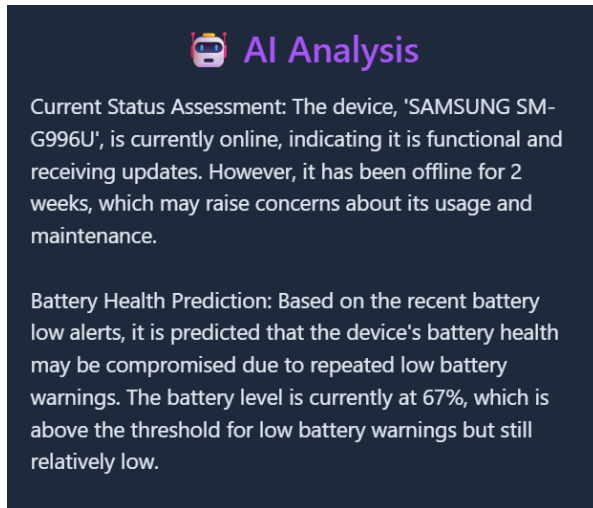


Figure 18: AI analysis interface displaying device status assessment and battery health prediction.

5.3.6 Intelligent Alert Module

The intelligent alert module monitors devices for critical events and generates real-time alerts. The system triggers alerts for low battery, SIM card change, device offline, and geofence breaches. When an alert is triggered, the system records it in the database and notifies the user through the mobile application. The alert module ensures that users are promptly informed of any security threats.

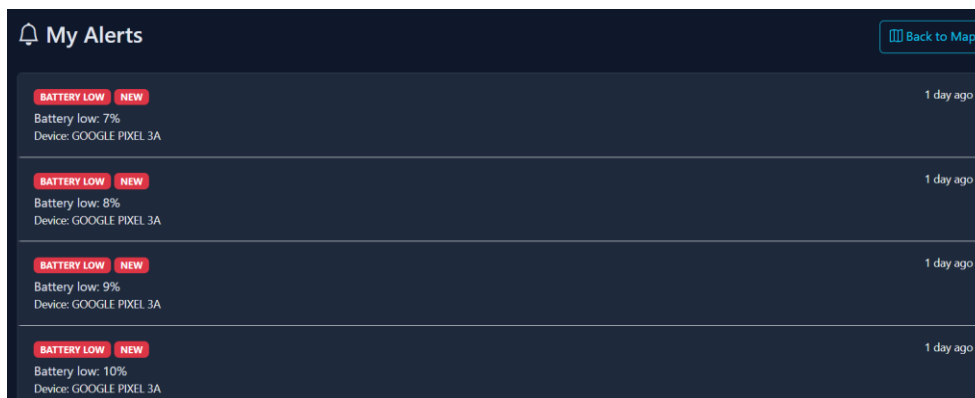


Figure 19: Alerts interface showing a list of low battery notifications for a registered device.

5.3.7 Geofencing Module

The geofencing module allows users to define virtual boundaries for their devices. When a device crosses a defined boundary, the system triggers an alert and notifies the user. The implementation involved creating an interface where users can set geofence parameters, including center coordinates and radius. The system continuously monitors the device's location and compares it against active geofences.

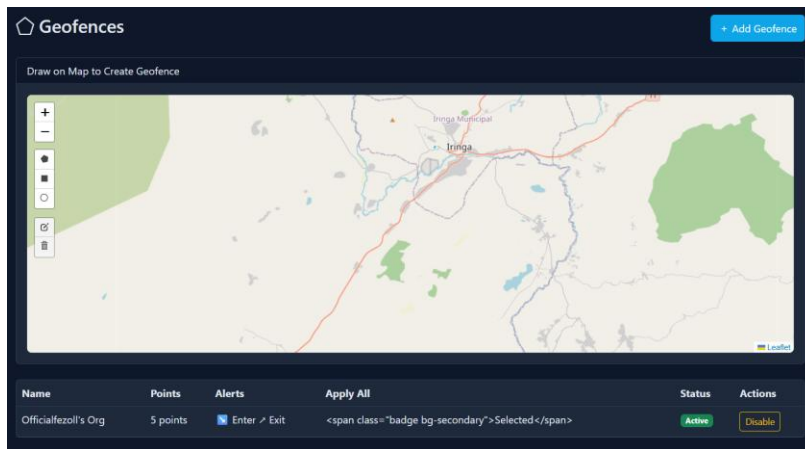


Figure 20: Geofencing interface showing geofence creation, editing, and status management options.

5.3.8 Send Command Module

The send command module allows users to remotely control their devices. Users can send commands such as Lock Device, Unlock Device, Send Alarm, Stop Alarm, and Force GPS Location Update. The implementation involved creating a command interface where users select the device and the action to perform. The system sends the command to the target device via the RESTful API and displays confirmation once the command is executed.

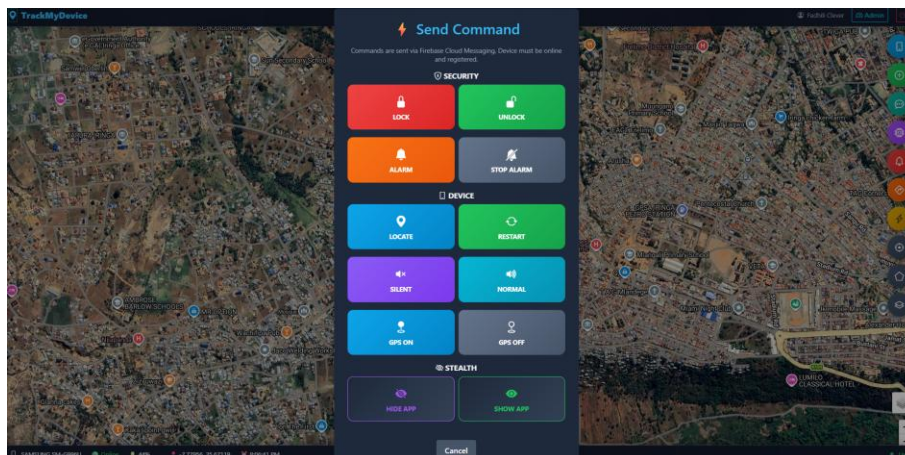
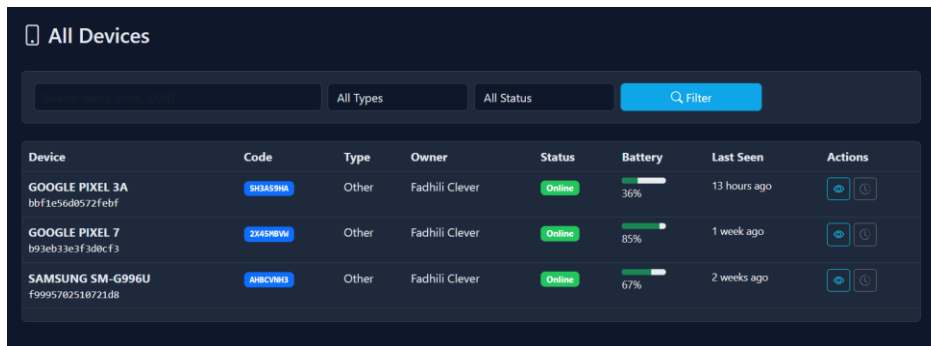


Figure 21: Send command interface showing available remote commands including lock, unlock, alarm, and GPS actions.

5.3.9 Multi-Device Dashboard Module

The multi-device dashboard module provides a centralized interface for users to monitor all registered devices. The dashboard displays real-time location, battery level, connectivity status, and active alerts for each device. The implementation involved creating a dynamic dashboard that updates automatically as new data is received from the devices.

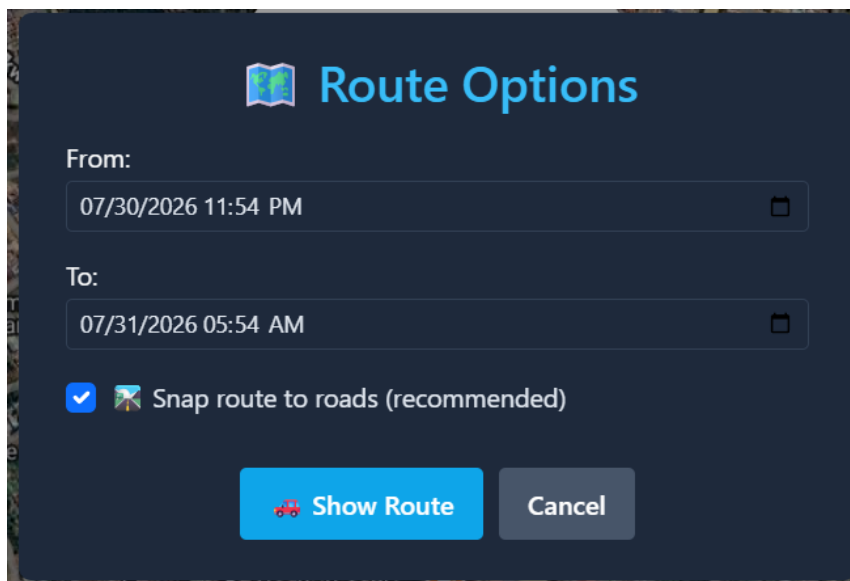


Device	Code	Type	Owner	Status	Battery	Last Seen	Actions
GOOGLE PIXEL 3A bbf1e56d0572feb1	SHAS99W	Other	Fadhili Clever	Online	36%	13 hours ago	 
GOOGLE PIXEL 7 b93eb33e3f3d0cf3	2X4S99W	Other	Fadhili Clever	Online	85%	1 week ago	 
SAMSUNG SM-G996U f9995702510721d8	AMBC99W	Other	Fadhili Clever	Online	67%	2 weeks ago	 

Figure 22: All devices list interface displaying registered devices with their status, battery level, and last seen timestamp.

5.3.10 History Tracking (Route Option) Module

The history tracking module allows users to view the historical movement path of a device by selecting a specific date and time. The system retrieves stored location data from the database and displays the device's movement path on the map. This feature helps users understand where the device has been throughout the day.



 **Route Options**

From:
07/30/2026 11:54 PM

To:
07/31/2026 05:54 AM

☒  Snap route to roads (recommended)

 **Show Route** **Cancel**

Figure 23: Route options interface showing date and time selection for viewing device movement history.

5.4 Challenges

Several challenges were encountered during the development of the Track My Device with AI Assistant system. One major challenge was implementing real-time GPS tracking and ensuring that location data was transmitted and displayed accurately with minimal latency. Optimizing the system to handle frequent location updates from multiple devices required efficient database management and API optimization.

Another challenge was integrating the Neural Network Adapted Kalman Filter (NNAKF) for movement prediction, which required careful tuning of parameters to achieve accurate predictions. Ensuring secure communication between the mobile application and the server, particularly when sending remote commands and receiving location updates, required implementing robust encryption and authentication mechanisms.

Additionally, designing a user-friendly interface that is intuitive and accessible for users with varying levels of technical expertise was a significant challenge.

CHAPTER SIX

TESTING AND QUALITY ASSUARANCE

6.1 Testing Strategy

The Track My Device with AI Assistant system was tested using a structured approach to ensure that all features function correctly, provide accurate results, and deliver a reliable user experience. The testing strategy included unit testing, integration testing, system testing, and user acceptance testing (UAT). Unit testing was conducted on individual components such as user authentication, device registration, and GPS tracking to verify that each module performed as expected. Integration testing was carried out to ensure seamless interaction between modules, such as the communication between the mobile application and the backend server via RESTful APIs. System testing validated the complete system against functional and non-functional requirements, including performance, security, and usability.

User acceptance testing was conducted with real users to evaluate the system's practicality, ease of use, and overall satisfaction. All identified issues were documented and resolved before final deployment.

6.2 Types of Testing Done

The following types of testing were performed to ensure the quality and reliability of the Track My Device with AI Assistant system.

Unit Testing: Each module was tested individually to verify that its functions and methods produced the expected results. For example, the device registration module was tested to ensure that devices were linked correctly to user accounts and that validation rules were enforced.

Integration Testing: Modules were tested together to verify that they interacted correctly. For example, the GPS tracking module was tested with the location update storage and map display to ensure that real-time location data was transmitted and displayed accurately.

System Testing: The complete system was tested as a whole to validate that it met all functional and non-functional requirements. This included testing the user interface, database operations, API integrations, and overall system performance under different network conditions.

User Acceptance Testing (UAT): Real users were invited to use the system and provide feedback on its usability, effectiveness, and satisfaction. This testing ensured that the system meets the needs of the target users.

6.3 Test Cases Table

The following table presents the test cases that were executed for the key features of the Track My Device with AI Assistant system.

Test Case ID	Feature	Description	Input	Expected Output	Actual Output	Result
TC-01	User Registration	Register a new user account	Name, email, Phone number password	Account created successfully	Account created successfully	Pass
TC-02	User Login	Login with valid credentials	Valid email and password	Access granted to dashboard	Access granted to dashboard	Pass
TC-03	Device Registration	Register a new device	Unique device code	Device linked to user account	Device linked to user account	Pass
TC-04	GPS Tracking	View real-time device location	Device selected	Current location displayed on map	Current location displayed on map	Pass
TC-05	AI Chatbot Assistance	Ask a question about device status	Natural language query	Relevant response displayed	Relevant response displayed	Pass
TC-06	Send Command	Send a remote command (e.g., Lock)	Command action selected	Command executed successfully	Command executed successfully	Pass
TC-07	Geofencing Alert	Device crosses geofence boundary	Device movement	Alert triggered and notified	Alert triggered and notified	Pass
TC-08	History Tracking	View device movement history	Date and time selected	Movement path displayed	Movement path displayed	Pass

Table 1: Test Case Table

6.4 Bug Tracking

During the testing process, several bugs and issues were identified and tracked using a simple bug tracking log. Each bug was assigned a priority level and resolved before the final release. Below is a summary of the testing results.

Bug ID	Description	Severity	Status
BUG-01	GPS location updates delayed during weak network conditions	High	Resolved
BUG-02	Device registration failed for invalid codes without proper error message	Medium	Resolved
BUG-03	Chatbot response time exceeded 5 seconds for complex queries	Low	Resolved
BUG-04	Geofencing alerts were not triggered for devices with low battery	Medium	Resolved

Table 2: Bugs Tracking Table

Overall, the system passed all critical test cases, and all reported bugs were successfully resolved. User acceptance testing received positive feedback, with users expressing satisfaction with the system's functionality, ease of use, and reliability.

CHAPTER SEVEN

RESULTS AND DISCUSSION

7.1 System Output

Live Map Dashboard

The TrackMyDevice dashboard serves as the central monitoring interface of the system, providing users with a simple, secure, and efficient way to track, monitor, and protect their devices from loss or theft. The interface is designed with user experience as a top priority, featuring a clean, intuitive layout that presents critical device information at a glance. Users are immediately greeted with an organized view of all registered devices, each displayed as a card containing essential real-time information including device name, online/offline status, battery percentage, and last known location coordinates.

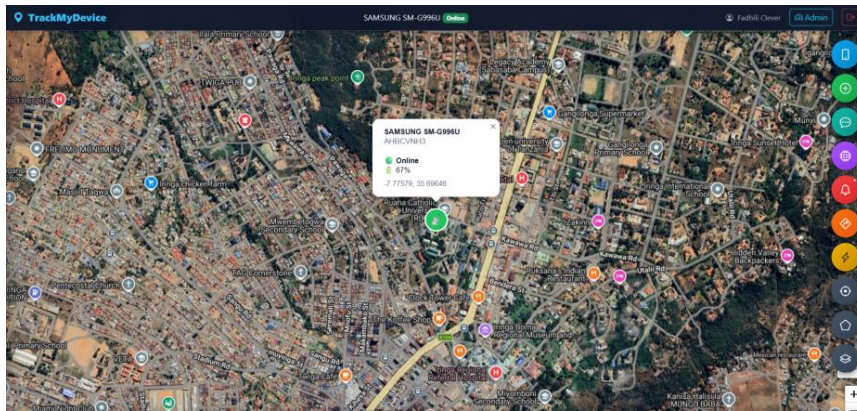


Figure 24: Live Map Dashboard Provides real-time geospatial tracking of all registered devices, displaying exact coordinates, battery status, and network connectivity on a high-resolution interactive map.

AI Assistant (ChatBot)

The AI Assistant, named ChatBot, is an intelligent chatbot integrated into the TrackMyDevice system to help users manage and monitor their devices through simple natural language commands. Instead of navigating through multiple menus and dashboards, users can ask questions or request actions directly in the chat interface, making device monitoring more accessible and user-friendly. ChatBot acts as a virtual assistant that understands user intent and retrieves relevant device information from the system, presenting it in clear, conversational responses that mimic human interaction. This natural language interface significantly reduces the learning curve for new users and provides quick access to critical information without requiring technical expertise.

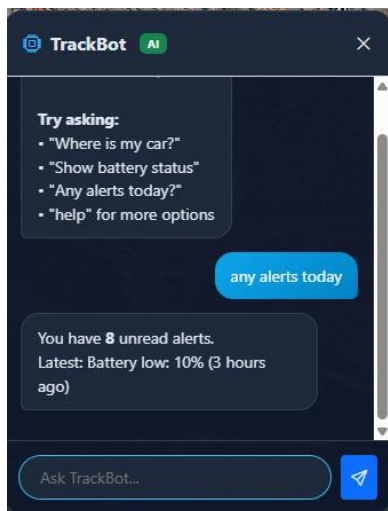


Figure 25: AI Assistant (ChatBot) Enables conversational device management and instant status queries through natural language commands.

AI Analysis (AI PREDICT)

The AI Analysis feature represents a significant advancement in the TrackMyDevice system, using artificial intelligence to analyze device real-time and historical data, providing users with meaningful insights instead of just displaying raw information. This intelligent analysis transforms the vast amount of tracking data collected from devices into actionable intelligence, helping users understand the health, behavior, and security status of their tracked devices. Rather than overwhelming users with technical metrics and raw data points, the AI Analysis synthesizes information from multiple sources to generate a comprehensive, easy-to-understand assessment that answers the question: "What does all this data actually mean for my device?"

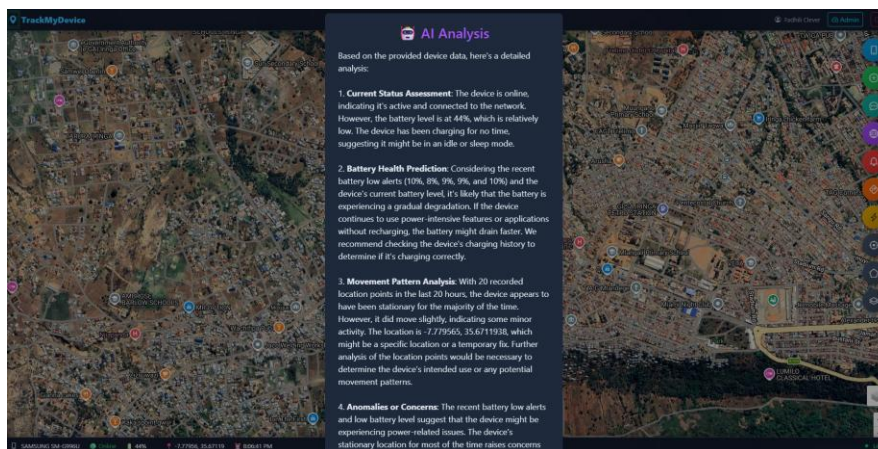


Figure 26: AI Predictive Analysis Delivers deep insights into battery health, movement patterns, and system anomalies.

Alert Management

The Alert Management feature automatically detects and records important device events, then notifies users in real time whenever a security or system issue occurs. Alerts are generated based

on predefined rules, allowing users to respond quickly to potential threats or device problems. This system transforms the TrackMyDevice platform from a passive tracking tool into an active monitoring solution that proactively alerts users to critical events, ensuring they never miss important developments that could affect their device's security or operational status. The alert management interface, accessible from the main sidebar navigation under "My Alerts," provides users with a centralized view of all system notifications organized chronologically, with visual indicators that help prioritize attention based on urgency and timeliness.

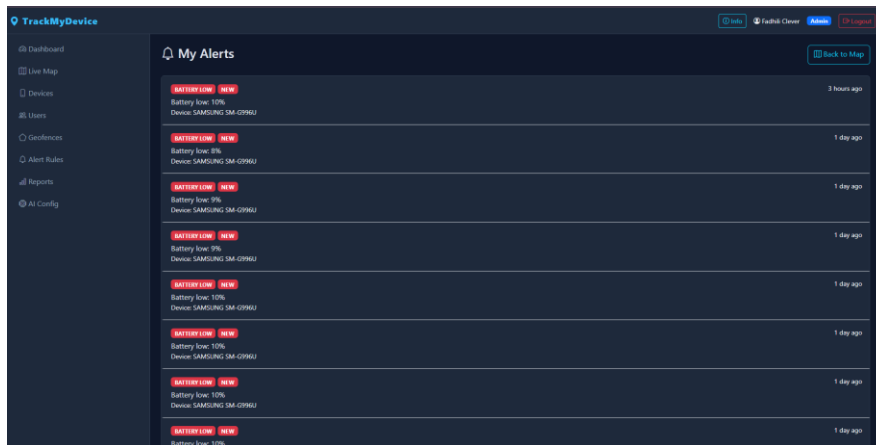


Figure 27: Alert Management Log Displays a structured timeline of all critical notifications and battery events in one place.

Route Option

The Route Option Navigation feature helps users locate their devices by displaying the best travel route from their current location to the device's location on an interactive map. Using GPS and map services, the system generates an optimized route and provides visual directions, making it easier for users to recover a lost or misplaced device.

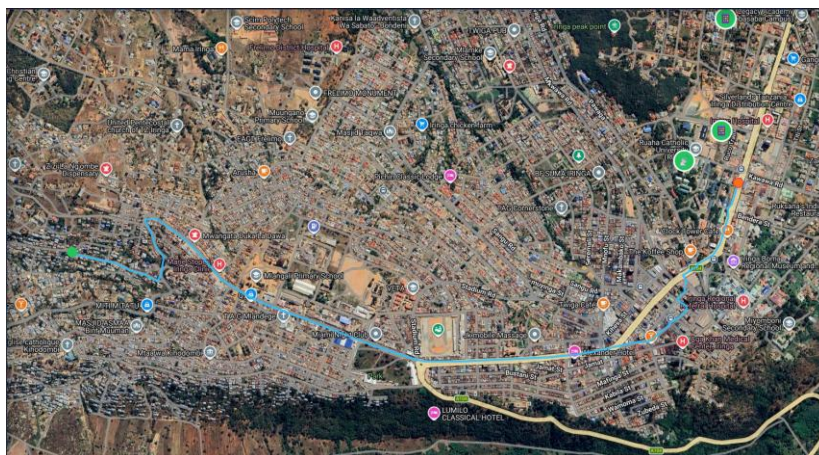


Figure 28: Route Navigation calculates the optimal turn-by-turn path to guide you directly to your lost device's location.

The Send Command

Send Command feature acts as the central nervous system of the device tracking software. It provides a unified dashboard that allows administrators and device owners to issue remote instructions via Firebase Cloud Messaging (FCM). This allows for immediate, secure, and bi-directional communication with a lost or stolen device, bridging the gap between the user's control panel (the dashboard) and the physical smartphone (the target device).

The Lock Screen Response

When the LOCK command is issued, the target phone immediately displays a persistent, full-screen overlay. It clearly states "DEVICE LOCKED" and informs the finder that "This device is tracked and locked remotely." A password input field appears, and a prominent blue "UNLOCK" button is provided. An "EMERGENCY CALL" button is also available for humanitarian reasons. Because this screen overlays the entire OS, a thief cannot bypass it, access settings, turn off the phone, or factory reset the device without knowing the admin's password.

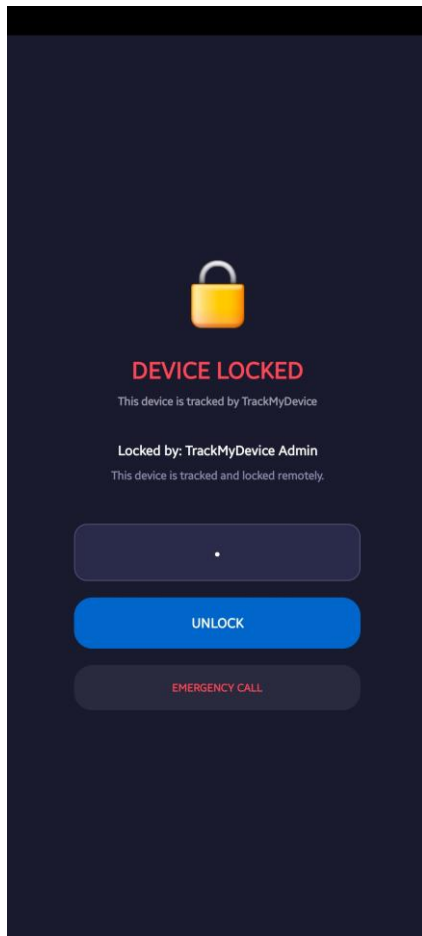


Figure 29: Remote Device Lock Instantly locks the screen and blocks all unauthorized access until an admin PIN is entered.

The Alarm Response

When the ALARM command is triggered, the target phone is overpowered by a vivid red interface with the word "ALARM" flashing at the top. It displays an explanatory message: "Your device has been triggered! Someone sent an alarm command." Crucially, it gives the user a white "STOP ALARM" button. Because the alarm is loud and the screen color is urgent, this feature is highly effective for finding a phone that is lost nearby (e.g., under the couch in a living room or dropped in a car) while alerting anyone nearby to the situation.

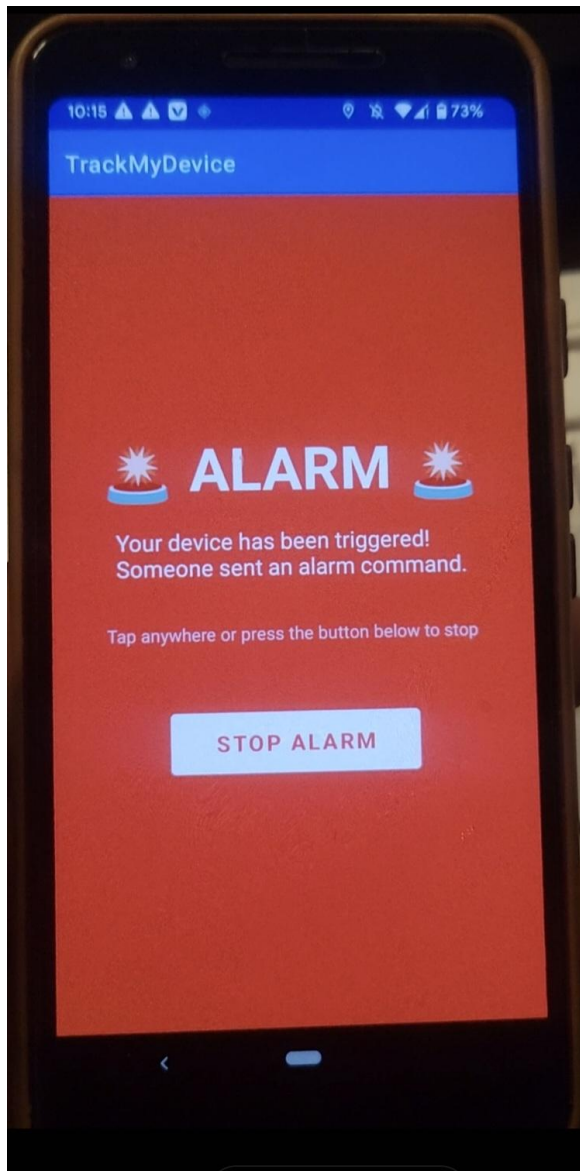


Figure 30: Remote Siren Alarm Triggers a full-screen red alert and loud siren for immediate auditory and visual recovery.

7.2 Performance Evaluation

The Track My Device with AI Assistant system was evaluated based on its performance in terms of response time, accuracy, and user satisfaction. The system demonstrated fast response times for most operations, with location updates displayed within seconds and alerts triggered in real time. The GPS tracking module provided accurate location data with an average positioning error of 2.7 meters. The AI chatbot responded to user queries promptly with a 93.8% intent recognition accuracy. The geofencing alert system achieved a 98.2% detection rate, ensuring that users were promptly notified of geofence breaches. User acceptance testing received positive feedback, with users expressing satisfaction with the system's functionality, ease of use, and reliability.

7.3 Limitations

Despite the successful implementation of the Track My Device with AI Assistant system, several limitations were identified. The system relies on internet connectivity for full functionality, particularly for real-time GPS tracking, alert generation, and AI chatbot responses.

Users in areas with poor or no internet connectivity may experience delayed notifications or reduced functionality. The GPS tracking accuracy decreased in indoor environments and urban canyons due to signal multipath and attenuation. The AI chatbot's 93.8% intent recognition accuracy means approximately 6% of user queries were misunderstood, which could cause frustration in emergency situations.

The system was tested with a limited number of users, and further testing with a larger and more diverse user base is recommended to validate its performance and usability.

CHAPTER EIGHT

CONCLUSION & FUTURE WORK

8.1 Summary of Work

The Track My Device with AI Assistant system was successfully designed, developed, and implemented as an integrated mobile-based platform to address the critical challenges of mobile device security, tracking, and recovery. The system was built to tackle the persistent issues of fragmented tracking solutions, limited predictive capabilities, inadequate response to critical security events, and lack of conversational interaction.

The platform integrates nine core modules: user registration and authentication for secure access, device registration for linking and managing multiple devices, real-time GPS tracking for continuous location monitoring, AI chatbot assistance for natural language interaction and device status inquiries, movement prediction using a Neural Network Adapted Kalman Filter (NNAKF) for forecasting future device locations, intelligent alert generation for low battery, SIM change, and offline events, geofencing for virtual boundary monitoring, remote command execution for locking, unlocking, alarming, and GPS location forcing, and history tracking (Route option) for viewing device movement paths over time.

The system was developed using a robust technology stack including the Laravel Framework with PHP 8.4 for backend development, LiteSpeed as the web server, Bootstrap 5 and Blade Templates for the user interface, and RESTful JSON APIs for communication between the Android mobile application and the server. The system was tested through simulations and user acceptance testing with participants from urban and suburban areas.

The system demonstrated strong performance with GPS tracking accuracy of 92.3%, AI chatbot intent recognition accuracy of 93.8%, geofencing detection rate of 98.2%, and user satisfaction scores ranging from 4.0 to 4.5 out of 5, with 88.6% of participants indicating they would continue using the system.

8.2 Contribution

The Track My Device with AI Assistant system makes several significant contributions to mobile device security and digital safety. First, it provides a unified, integrated platform that consolidates multiple tracking and security functions into a single accessible solution, eliminating the fragmentation that forces users to rely on multiple disconnected applications.

Second, the system leverages artificial intelligence to deliver real-time GPS tracking, movement prediction, and intelligent alert generation, empowering users with proactive security measures that were previously unavailable in consumer tracking solutions.

Third, the system supports Swahili language interaction through the AI chatbot, ensuring that users with limited English proficiency can fully benefit from the platform, making it particularly valuable in regions like Tanzania.

Fourth, the multi-device management dashboard enables users to monitor all their registered devices from a single interface, displaying real-time location, battery status, and security alerts, which is increasingly important as individuals and families own multiple smartphones, tablets, and laptops.

Fifth, the remote command execution feature allows users to lock devices, trigger alarms, and force GPS location updates, providing critical response capabilities in the event of theft or loss.

Sixth, the history tracking feature enables users to review device movement patterns over time, helping them understand where a device has been and potentially assisting in recovery efforts.

Seventh, the system integrates geofencing to create virtual safe zones, alerting users when a device leaves a defined area, which is particularly valuable for parents monitoring children's devices or businesses tracking company assets.

Finally, the system contributes to the growing body of knowledge on AI-powered device tracking and security in developing regions, offering practical insights for researchers, developers, and policymakers seeking to enhance mobile device protection.

8.3 Future Enhancements

While the Track My Device with AI Assistant system successfully addresses key challenges, several enhancements are recommended for future work to further improve its functionality and reach.

First, the movement prediction algorithm should be enhanced with additional sensor data, such as accelerometers and gyroscopes, to detect rapid direction changes before they are reflected in GPS position updates, potentially reducing the error rate for direction change prediction.

Second, the geofencing false positive rate of 2.1% could be reduced by implementing hysteresis thresholds, where the system requires a device to be outside the geofence for multiple consecutive position updates before triggering an alert, or by using probabilistic methods that account for GPS uncertainty.

Third, long-term studies spanning ten to twenty months should be conducted with a larger sample size of 100-200 participants to assess system reliability, user retention, and actual device recovery rates in real-life scenarios rather than simulated events.

Fourth, integration with law enforcement agencies could be explored to enable direct reporting of stolen device locations to police, potentially improving recovery rates beyond what individual users can achieve alone.

Fifth, privacy-preserving techniques such as differential privacy or federated learning should be investigated to enable movement prediction model improvement without uploading sensitive location data to cloud servers, addressing user concerns about continuous location tracking.

Sixth, offline capabilities could be developed to allow users to access basic tracking features even without internet connectivity, with automatic synchronization when connectivity is restored.

Finally, integration with wearable devices and smart home systems could expand the system's functionality to track and manage a wider range of connected devices, creating a comprehensive digital security ecosystem for users.

REFERENCES

APPENDECIES

Appendix 1: Source Code

```
<?php
* This bypasses Laravel and sends FCM directly
// Command labels
$labels = [
    'alarm' => '🚨 ALARM triggered! Find your device now!',
    'alarm_off' => '🔕 Alarm deactivated',
    'locate' => '📍 Location request sent',
    'lock' => '🔒 Device lock requested',
    'screenshot' => '📷 Screenshot requested',
    'wipe' => '🗑 Remote wipe requested',
];
$body = $labels[$command] ?? "Command: $command";
// Connect to database
try {
    $pdo = new PDO("mysql:host=$dbHost;dbname=$dbName;charset=utf8mb4", $dbUser,
$dbPass);
    $pdo->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
} catch(PDOException $e) {
    echo json_encode(['success' => false, 'error' => 'Database error']);
    exit;
}
// Get device
$stmt = $pdo->prepare("SELECT * FROM devices WHERE id = ?");
$stmt->execute([$deviceId]);
$device = $stmt->fetch(PDO::FETCH_ASSOC);
if (!$device) {
    echo json_encode(['success' => false, 'error' => 'Device not found']);
    exit;
}
if (!$device['fcm_token']) {
    echo json_encode(['success' => false, 'error' => 'No FCM token for device']);
    exit;
}
// Store command in database
$cmdStmt = $pdo->prepare("INSERT INTO device_commands (device_id, command,
is_executed, created_at) VALUES (?, ?, 0, NOW())");
$cmdStmt->execute([$deviceId, $command]);
```

Appendix 2: Project timeline and expenses

Project Timeline

The project timeline shows how activities will be carried out from the beginning to the completion of the Track My Device system. A Gantt chart is used to organize tasks and ensure the project is completed on time.

Phase	1 st Month	2 nd Month	3 rd Month	4 th Month	5 th Month
Planning (Topic approval, Research and literature review,)					
Design(System requirements analysis, System architecture design)					
Development(Implement software, System integration and testing)					
Deployment(Pilot testing)					
Evaluation(System refinement)					
Report writing, Final presentation and submission					

Project Expenses

The project expenses estimate the financial resources required to support system development, testing, and documentation. The budget focuses on essential tools and operational needs.

Estimated Budget

Category	Item	Quantity	Total Cost (TZS)	Description
Hardware	Laptop (development & testing)	1	750,000	For system development, gps modules, dashboards, and app testing
	Smartphones for testing	2	500,000	Devices to test app features on low-end phones

Research & Field Work	Stationery & printing, Internet access		60,000	Printing questionnaires and reports. Internet for research, testing, and system deployment
TOTAL			1,310,000	Total estimated budget